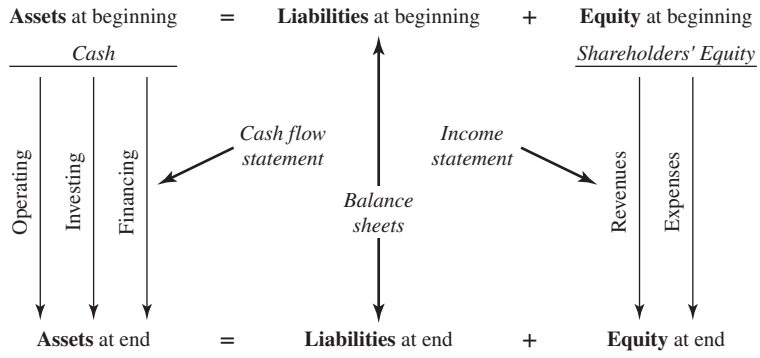
Higgins

Jennifer L. Koski_ Robert C. Higgins_ Todd Mitton - Analysis

- Balance sheet and income statement: Ch. 1, §1.3–1.4 (p. 9–16)
- Sources and uses; cash flow statement: Ch. 1, §1.5–1.6 (p. 17–22)
- ROE = margin × turnover × leverage; turnover and leverage ratios: Ch. 2, §2.2 (p. 38–53)
- ROE risk problem and ROIC: Ch. 2, §2.3.1–2.3.2 (p. 54–56)

FIGURE 1.2 **Ties among Financial Statements**



The Balance Sheet

POLARIS

See **polaris.com**. Follow Company > Investors > Financials & Filings for financial statements.

We will now take a more in-depth look at each of the key financial statements in turn, beginning with the balance sheet. To illustrate the techniques and concepts presented throughout the book, I will refer whenever possible to Polaris Inc., the self-styled "Global leader in Powersports" including snowmobiles, all-terrain vehicles (ATVs), motorcycles, and boats. In addition to the United States, Polaris has manufacturing operations in China, France, Mexico, and Poland. It has offices in 14 countries and sells its products in over 100 countries, although 80 percent of sales originate in the United States. The company's focus, as embodied in its marketing tagline "Think Outside" is to create products that enhance the experience of outdoor enthusiasts.

Headquartered in Medina, Minnesota, a small town outside of Minneapolis, Polaris has $7 billion in annual sales, and its stock trades on the New York Stock Exchange. The firm was founded in 1954 as one of the pioneers of the snowmobile industry. In the 1980s, Polaris diversified into off-road vehicles when they produced the first American-made ATVs. In 2011, Polaris extended its reach into on-road vehicles with its acquisition of the legendary Indian Motorcycle company. Between 2018 and 2019, two more acquisitions added three well-known boat brands to Polaris's product portfolio. Polaris's recent ventures include forays into electric ATVs, electric motorcycles, and special-purpose electric automobiles. In 2020, 64 percent of Polaris's sales came from snowmobiles and off-road vehicles, with 9 percent coming from boats, 8 percent from motorcycles, and 19 percent from other products and services.

Table 1.2 presents Polaris's balance sheets for 2019 and 2020. If the precise meaning of every asset and liability category in Table 1.2 is not immediately apparent, be patient. We will discuss many of them in the following text. In addition, all of the accounting terms used appear in the glossary at the end of the book.

Polaris Inc.'s balance sheet equation for 2020 is

Assets	= Liabilities	+ Shareholders' equity
$4,922 million	= $3,488 million	+ $1,434 million

Current Assets and Liabilities

By convention, U.S. accountants list assets and liabilities on the balance sheet in order of decreasing liquidity, where liquidity refers to the speed with which an item can be converted to cash. Thus, cash, marketable securities, and accounts receivable appear at the top, while land, plant, and equipment are toward the bottom. Similarly on the liabilities side, short-term loans and accounts payable are toward the top, while shareholders' equity is at the bottom.

TABLE 1.2 **Polaris Inc. Balance Sheets ($ millions)***

Source: Polaris annual reports

	December 31		Change in Account
	2019	2020	
Assets			
Cash and cash equivalents	$ 157	$ 544	$ 387
Accounts receivable	190	257	67
Inventories	1,121	1,178	57
Other current assets	158	138	(20)
Total current assets	1,627	2,117	
Gross property, plant, and equipment	2,004	2,142	138
Less accumulated depreciation and amortization	994	1,128	134
Net property, plant, and equipment	1,010	1,014	4
Goodwill and other intangible assets	1,490	1,463	(27)
Other assets	303	327	24
Total assets	$ 4,431	$4,922	
Liabilities and Shareholders' Equity			
Debt due in one year	167	142	(25)
Accounts payable	450	782	332
Accrued liabilities	870	908	38
Other current liabilities	41	57	16
Total current liabilities	1,528	1,889	
Long-term debt	1,512	1,294	(218)
Other long-term liabilities	282	305	23
Total liabilities	3,322	3,488	
Common stock	1	1	
Additional paid-in capital	893	984	
Retained earnings	215	449	
Total shareholders' equity	1,108	1,434	326
Total liabilities and shareholders' equity	$ 4,431	$4,922	

*Totals may not add due to rounding.

Accountants also arbitrarily define any asset or liability that is expected to turn into cash within one year as *current* and all others assets and liabilities as *long term.* Inventory is a current asset because there is reason to believe it will be sold and will generate cash within one year. Accounts payable are short-term liabilities because they must be paid within one year. Note that over 40 percent of Polaris's assets are current, a fact we will say more about in the next chapter.

A Word to the Unwary

Nothing puts a damper on a good financial discussion (if such exists) faster than the suggestion that if a company is short of cash, it can always spend some of its shareholders' equity. Equity is on the liabilities side of the balance sheet, not the asset side. It represents owners' claims against existing assets. In other words, that money has already been spent.

Shareholders' Equity

The different accounts appearing in the shareholders' equity portion of the balance sheet can often be a source of confusion. Polaris has three equity accounts: common stock, additional paid-in capital, and retained earnings (see Table 1.2). Unless forced to do otherwise, my advice is to forget these distinctions. They keep accountants and attorneys employed, but seldom make much practical difference. As a first cut, just add up everything that is not an IOU and call it shareholders' equity.

The Income Statement

Polaris's income statements for 2019 and 2020 are presented in Table 1.3. Looking at Polaris's operating performance in 2020, the basic income statement relation appearing in Table 1.3 is:

Revenues	–	Expenses									=	Net income
Net sales	–	Cost of goods sold	–	Operating expenses	–	Nonoperating expenses	–	Taxes	=	Net income		
$7,108	–	$5,057	–	$1,460	–	$71	–	$107	=	$414		

Net income records the extent to which net sales generated during the accounting period exceeded expenses incurred in producing the sales. For variety, net income is also commonly referred to as *earnings* or *profits,* frequently with the word *net* stuck in front of them; net sales are often called *revenues* or *net revenues;* and cost of goods sold is sometimes labeled *cost of sales.* I have never found a meaningful distinction between these terms. Why so many words to say the same thing? My personal belief is that accountants are

TABLE 1.3 Polaris Inc. Income Statements ($ millions)*

Source: Polaris annual reports and Compustat

	January 1 to December 31	
	2019	**2020**
Net sales	$6,863	$7,108
Cost of goods sold	4,899	5,057
Gross profit	1,964	2,051
Selling, general, and administrative expenses	953	904
Research and development	293	296
Depreciation and amortization	235	261
Total operating expenses	1,481	1,460
Operating income	484	592
Interest expense	78	67
Other nonoperating expense (income)	(2)	4
Total nonoperating expenses	76	71
Income before income taxes	408	521
Provision for income taxes	84	107
Net income	$ 324	$ 414

*Totals may not add due to rounding.

so rule-bound in their calculations of the various amounts that their creativity runs a bit amok when it comes to naming them.

Income statements are commonly divided into operating and nonoperating segments. As the names imply, the operating segment reports the results of the company's major, ongoing activities, while the nonoperating segment summarizes all ancillary activities. In 2020, Polaris reported operating income of $592 million and nonoperating expenses of $71 million, consisting largely of interest expense.

Measuring Earnings

This is not the place for a detailed discussion of accounting. But because earnings, or lack of same, are a critical indicator of financial health, several technical details of earnings measurement deserve mention.

Accrual Accounting

The measurement of accounting earnings involves two steps: (1) identifying revenues for the period and (2) matching the corresponding costs to revenues. Looking at the first step, it is important to recognize that revenue is not the same as cash received. According to the *accrual principle* (a cruel principle?) of accounting, revenue is recognized as soon as "the effort required to generate the sale is substantially complete and there is a reasonable certainty that payment will be received." The accountant sees the timing of the actual

cash receipts as a mere technicality. For credit sales, the accrual principle means that revenue is recognized at the time of sale, not when the customer pays. This can result in a significant time lag between the generation of revenue and the receipt of cash. Looking at Polaris, we see that revenue in 2020 was $7,108 million, but accounts receivable increased by $67 million. We conclude that cash received from sales during 2020 was only $7,041 million ($7,108 – $67 million). The other $67 million still awaits collection.

Depreciation

Fixed assets and their associated depreciation present the accountant with a particularly challenging problem in matching. Suppose that in 2022, a company constructs, for $50 million, a new facility that has an expected productive life of 10 years. If the accountant assigns the entire cost of the facility to expenses in 2022, some weird results follow. Income in 2022 will appear depressed due to the $50 million expense, while income in the following nine years will look that much better as the new facility contributes to revenue but not to expenses. Thus, charging the full cost of a long-term asset to one year clearly distorts reported income.

The preferred approach is to spread the cost of the facility over its expected useful life in the form of depreciation. Because the only cash outlay associated with the facility occurs in 2022, the annual depreciation listed as a cost on the company's income statement is not a cash outflow. It is a *noncash charge* used to match the 2022 expenditure with resulting revenue. Said differently, depreciation is the allocation of past expenditures to future time periods to match revenues and expenses. A glance at Polaris's income statement reveals that, in 2020, the company included a $261 million noncash charge for depreciation and amortization among their operating expenses. Further along in the text, we will see that during the same year, the company spent $214 million acquiring new property, plant, and equipment.

To determine the amount of depreciation to take on a particular asset, three estimates are required: the asset's useful life, its salvage value, and the method of allocation to be employed. These estimates should be based on economic and engineering information, experience, and any other objective data about the asset's likely performance. Broadly speaking, there are two methods of allocating an asset's cost over its useful life. Under the *straight-line* method, the accountant depreciates the asset by a uniform amount each year. If an asset costs $50 million, has an expected useful life of 10 years, and has an estimated salvage value of $10 million, straight-line depreciation will be $4 million per year ([$50 million – $10 million]/10).

The second method of cost allocation is really a family of methods known as *accelerated depreciation.* Each technique charges more depreciation in the early years of the asset's life and correspondingly less in later years. Accelerated depreciation does not enable a company to take more depreciation in

total; rather, it alters the timing of the recognition. While the specifics of the various accelerated techniques need not detain us here, you should recognize that the life expectancy, the salvage value, and the allocation method a company uses can fundamentally affect reported earnings. In general, if a company is conservative and depreciates its assets rapidly, it will tend to understate current earnings, and vice versa.

Taxes

A second noteworthy feature of depreciation accounting involves taxes. Most U.S. companies, except very small ones, keep at least two sets of financial records: one for managing the company and reporting to shareholders, and another for determining the firm's tax bill. The objective of the first set is, or should be, to accurately portray the company's financial performance. The objective of the second set is much simpler: to minimize taxes. Forget objectivity and minimize taxes. These differing objectives mean the accounting principles used to construct the two sets of books differ substantially. Depreciation accounting is a case in point. Regardless of the method used to report to shareholders, company tax books will minimize current taxes by employing the most rapid method of depreciation over the shortest useful life the tax authorities allow. In the United States, tax depreciation is governed by a set of rules called the Modified Accelerated Cost Recovery System (MACRS). We will not delve into the scintillating details of MACRS here, but interested readers can refer to IRS Publication 946, a 113-page guide to all things MACRS.

This dual reporting means that actual cash payments to tax authorities usually differ from the provision for income taxes appearing on a company's income statement, sometimes trailing the provision and other times exceeding it. When this happens, the company will report the difference somewhere on its balance sheet. If the company pays less than it owes, it will have a liability with a name like "taxes payable," representing the money it must pay tax authorities in future years. In the meantime, this money can be used to finance the business. Tax deferral techniques create the equivalent of interest-free loans from the government. In Japan and other countries that do not allow the use of separate accounting techniques for tax and reporting purposes, these complications never arise. Polaris had over $30 million in taxes payable in 2020; in Table 1.2, this amount is included as part of the "other" current and long-term liabilities listed on their balance sheet.

Research and Marketing

Now that you understand how accountants use depreciation to spread the cost of long-lived assets over their useful lives to better match revenues and costs, you may think you also understand how they treat research and marketing expenses. Because research and development (R&D) and marketing outlays promise benefits over a number of future periods, it is only logical that an

accountant would show these expenditures as assets when they are incurred and then spread the costs over the assets' expected useful lives in the form of a noncash charge such as depreciation. This may be impeccable logic, but it isn't what accountants do, at least not in the United States. Because the magnitude and duration of the prospective payoffs from R&D and marketing expenditures are difficult to estimate, accountants typically duck the problem by forcing companies to record the entire expenditure as an operating cost in the year incurred. Thus, although Polaris's R&D outlays of $296 million in 2020 may have produced technical breakthroughs that will benefit the firm for decades to come, all of the costs must be shown on Polaris's income statement in 2020. The requirement that companies expense all research and marketing expenditures when incurred commonly understates the profitability of high-tech and high-marketing companies and complicates comparison of American companies with those in other nations that treat such expenditures more liberally.

Defining Earnings

Creditors and investors look to company earnings for help in answering two fundamental questions: How did the company do last period, and how might it do in the future? To answer the first question, it is important to use a broad-based measure of income that includes everything affecting the company's performance over the accounting period. However, to answer the second question, we want a narrower income measure that abstracts from all unusual, nonrecurring events to focus strictly on the company's steady state, or ongoing, performance.

The accounting profession and the Securities and Exchange Commission obligingly provide two such official measures, known as net income and operating income, and require companies to report them on their financial statements.

Net income, or net profit, is the proverbial "bottom line," defined as total revenue less total expenses.

Operating income is profit realized from day-to-day operations excluding taxes, interest income and expense, and what are known as extraordinary items. An extraordinary item is one that is both unusual in nature and infrequent in occurrence.

Managers and investment professionals often consider several other, unofficial measures of earnings. Here are two of the more popular ones:

EBIT (pronounced E-bit) is earnings before interest and taxes, a useful and widely used measure of a business's income before it is divided among creditors, owners, and the taxman. EBIT is sometimes used interchangeably with operating income.

EBITDA (pronounced E-bit-da) is earnings before interest, taxes, depreciation, and amortization. EBITDA has its uses in some industries, such as broadcasting, where depreciation charges may routinely overstate true economic depreciation. However, as Warren Buffett notes, treating EBITDA as equivalent to earnings is tantamount to saying that a business is the commercial equivalent of the pyramids—forever state-of-the-art, never needing to be replaced, improved, or refurbished. In Buffett's view, EBITDA is a number favored by investment bankers when they cannot justify a deal based on EBIT.

Sources and Uses Statements

Two very basic but valuable things to know about a company are where it gets its cash and how it spends the cash. At first blush, it might appear that the income statement will answer these questions because it records flows of resources over time. But further reflection will convince you that the income statement is deficient in two respects: It includes accruals that are not cash flows, and it lists only cash flows associated with the sale of goods or services during the accounting period. A host of other cash receipts and disbursements do not appear on the income statement. Thus, Polaris increased its investment in accounts receivable by $67 million in 2020 (Table 1.2) with little or no trace of this buildup on its income statement. Polaris also decreased long-term debt by $218 million with little effect on its income statement.

To gain a more accurate picture of where a company got its money and how it spent it, we need to look more closely at the balance sheet or, more precisely, two balance sheets. Use the following two-step procedure. First, place two balance sheets for different dates side by side, and note all of the changes in accounts that occurred over the period. The changes for Polaris in 2020 appear in the rightmost column of Table 1.2. Second, segregate the changes into those that generated cash and those that consumed cash. The result is a *sources and uses statement.*

Here are the guidelines for distinguishing between a source and a use of cash:

- *A company generates cash in two ways: by reducing an asset or by increasing a liability.* The sale of used equipment, the liquidation of inventories, and the reduction of accounts receivable are all reductions in asset accounts and are all sources of cash to the company. On the liabilities side of the balance sheet, an increase in a bank loan and the sale of common stock are increases in liabilities, which again generate cash.
- *A company also uses cash in two ways: to increase an asset account or to reduce a liability account.* Adding to inventories or accounts receivable and building a new plant all increase assets and all use cash. Conversely, the repayment of a bank loan, the reduction of accounts payable, and an operating loss all reduce liabilities and all use cash.

Because it is difficult to spend money you don't have, total uses of cash over an accounting period must equal total sources.

Table 1.4 presents a 2020 sources and uses statement for Polaris. The largest source for Polaris is an increase in accounts payable of $332 million, meaning that Polaris has preserved cash by not paying its suppliers

TABLE 1.4 Polaris Inc. Sources and Uses Statement, 2020 ($ millions)

Source: Polaris annual reports and Compustat

Sources	
Reduction in other current assets	20
Reduction in goodwill and intangible assets	27
Increase in accounts payable	332
Increase in accrued liabilities	38
Increase in other current liabilities	16
Increase in other long-term liabilities	23
Increase in total shareholders' equity	326
Total sources	$782
Uses	
Increase in cash and cash equivalents	387
Increase in accounts receivable	67
Increase in inventories	57
Increase in net property, plant, and equipment	4
Increase in other assets	24
Decrease in debt due in one year	25
Decrease in long-term debt	218
Total uses	$782

immediately. The second largest source is an increase in shareholders' equity of $326 million, the bulk of which comes from profits earned by Polaris during the year. What has Polaris done with this cash? The two biggest uses are $218 million that has gone to pay down long-term debt, and $387 million that has been stashed in Polaris's cash account.

The Two-Finger Approach

I personally do not spend a lot of time constructing sources and uses statements. It might be instructive to go through the exercise once or twice just to convince yourself that sources really do equal uses. But once beyond this point, I recommend using a "two-finger approach." Put the two balance sheets side by side, and quickly run any two fingers down the columns in search of big changes. In the case of Polaris, this should enable you to quickly observe that Polaris used increases in equity and accounts payable primarily to pay down debt and build up its cash account. In 30 seconds or less, you have the essence of a sources and uses analysis and are free to move on to more stimulating activities. The other changes are largely window dressing of more interest to accountants than to managers.

How Can an Increase in Cash Be a Use of Cash?

One potential source of confusion in Table 1.4 is that the increase in cash and cash equivalents in 2020 appears as a use of cash. How can an increase in cash be a use of cash? Simple. It is the same as when you deposit money into your checking account. You increase your bank balance but have less cash on hand to spend. Conversely, a withdrawal from your bank account decreases your balance but increases spendable cash in your pocket.

The Cash Flow Statement

Identifying a company's principal sources and uses of cash is a useful skill in its own right. It is also an excellent starting point for considering the cash flow statement, the third major component of financial statements along with the income statement and the balance sheet.

In essence, a cash flow statement just expands and rearranges the sources and uses statement, placing each source or use into one of three broad categories. The categories and their values for Polaris in 2020 are as follows:

Category	Source (or Use) of Cash ($ millions)
1. Cash flows from operating activities	$945
2. Cash flows from investing activities	($151)
3. Cash flows from financing activities	($407)

Double-entry bookkeeping guarantees that the sum of the cash flows in these three categories equals the change in cash balances over the accounting period.

Table 1.5 presents a complete cash flow statement for Polaris in 2020. The first category, "cash flows from operating activities," can be thought of as a rearrangement of Polaris's financial statements to eliminate the effects of accrual accounting on net income. First, we add all noncash charges, such as depreciation and amortization, back to net income, recognizing that these charges did not entail any cash outflow. Then, we add the changes in current assets and liabilities to net income, acknowledging, for instance, that some sales did not increase cash because customers had not yet paid, while some expenses did not reduce cash because the company had not yet paid. Changes in other current assets and liabilities, such as inventories, appear here because the accountant, following the matching principle, ignored these cash flows when calculating net income.

If cash flow statements were just a reshuffling of sources and uses statements, as many textbook examples suggest, they would be redundant, for a reader could make his own in a matter of minutes. A chief attraction of

TABLE 1.5 Polaris Inc. Cash Flow Statement, 2020 ($ millions)*

Source: Polaris annual report and Compustat

Cash Flows from Operating Activities	
Net income	$ 414
Adjustments to reconcile net income to net cash provided by operating activities:	
Depreciation and amortization	261
Noncash compensation	65
Other	(100)
Changes in assets and liabilities:	
Accounts receivable	(56)
Inventories	(45)
Accounts payable	327
Other	80
Net cash provided by (used in) operating activities	945
Cash Flows from Investing Activities	
Capital expenditures	(214)
Other investing activities	63
Net cash provided by (used in) investing activities	(151)
Cash Flows from Financing Activities	
Net borrowing (repayment) of long-term debt	(246)
Net issuance (repurchase) of common stock	(17)
Dividends paid	(153)
Other	9
Net cash provided by (used in) financing activities	(407)
Net increase (decrease) in cash	387
Cash at beginning of year	157
Cash at end of year	$544

*Totals may not add due to rounding.

cash flow statements is that companies reorganize their cash flows into new and sometimes revealing categories. To illustrate, Polaris's cash flow statement in Table 1.5 reveals that during 2020 it paid dividends of $153 million, repurchased $17 million of its common stock (net of new stock issued), and invested $214 million in new capital expenditures. This is the only place in its financial statements where these basic activities are even mentioned.

A second attraction of a cash flow statement is that it casts a welcome light on firm solvency by highlighting the extent to which operations are generating or consuming cash. Polaris's cash flow statement in 2020 indicates that cash flow from operating activities more than doubled Polaris's net income for the year. One reason for the difference is that the income statement includes a $261 million noncash charge for depreciation. An even larger amount comes from the increase of over $300 million in accounts payable.

Why Are the Numbers Different?

Polaris's sources and uses statement in Table 1.4 tells us that accounts receivable rose $67 million in 2020 (a use of cash), yet its cash flow statement in Table 1.5 says that accounts receivable increased $56 million over the same period. Nor is this an isolated example. Many of the apparently identical quantities differ from one statement to the other. Why the difference?

Here are two possible answers. Companies often divide changes in current assets and liabilities into two parts: those attributable to existing activities, and those due to newly acquired businesses, with the first appearing in cash flows from operating activities and the second in cash flows from investing activities. By pushing uses of cash into investing activities as much as possible, a company enhances its recorded cash generated by operating activities—an appealing outcome. The second answer involves exchange rates. Companies like Polaris have assets and liabilities of various types scattered all over the world. To construct a consolidated balance sheet, accountants translate a company's foreign-denominated accounts into U.S. dollars at the then prevailing exchange rates. As a result, the balance sheet changes we observe on consolidated statements are due at least in part to changing currency values. However, because the currency-induced changes are not cash flows until the assets or liabilities are brought home, accountants omit them from the numbers appearing on its cash flow statement.

Are these answers complicated? Yes. Do the manipulations described add to our understanding of Polaris's performance? I doubt it.

Another noteworthy entry on Polaris's cash flow statement is "noncash compensation," which contributed $65 million to cash flow from operations in 2020. After a long and bitter battle among businesses, Congress, and accounting regulators, employee stock options are now, correctly, classified as an expense in the U.S. However, they are not a cash flow, because no cash exchanges hands when a company grants stock options to an employee. So they too must be added back to net income when calculating cash flow from operations. If you are wondering how stock options can be an expense if the company doesn't pay any cash to anyone, the answer is that they are a cost to shareholders, who see their ownership percentage diluted as employees acquire shares without paying full value for them.

Some analysts maintain that net cash provided by operating activities, appearing on the cash flow statement, is a more reliable indicator of firm performance than net income. They argue that because net income depends on myriad estimates, allocations, and approximations, devious managers can easily manipulate it. Numbers appearing on a company's cash flow statement, on the other hand, record the actual movement of cash, and are thus seen to be more objective measures of performance.

There is certainly some merit to this view, but it also has two problems. First, low or even negative net cash provided by operating activities does not necessarily indicate poor performance. Rapidly growing businesses in particular must customarily invest in current assets, such as accounts receivable

What Is Cash Flow?

So many conflicting definitions of *cash flow* exist today that the term has almost lost its meaning. At one level, cash flow is very simple. It is the movement of money into or out of a cash account over a period of time. The problem arises when we try to be more specific. Here are four common definitions of cash flow you are likely to encounter.

$$\text{Net cash flow} = \text{Net income} + \text{Noncash items}$$

Often known in investment circles as cash earnings, net cash flow is intended to measure the cash a business generates, as distinct from the earnings—a laudable objective. Applying the preceding formula to Polaris's 2020 figures (Table 1.5), its net cash flow was $640 million, equal to net income plus depreciation, noncash compensation, and other noncash adjustments.

A problem with net cash flow as a measure of cash generation is that it implicitly assumes a business's current assets and liabilities are either unrelated to operations or do not change over time. In Polaris's case, the cash flow statement reveals that changes in a number of current assets and liabilities produced $306 million in cash. A more inclusive measure of cash generation is therefore cash flow from operating activities as it appears on the cash flow statement.

$$\begin{aligned}\text{Cash flow from operating activities} = {} & \text{Net cash flow} \\ & \pm \text{Changes in current assets and liabilities}\end{aligned}$$

A third, even more inclusive measure of cash flow, popular among finance specialists is

$$\text{Free cash flow} = \begin{array}{l}\text{Total cash available for distribution to owners and creditors} \\ \text{after funding all worthwhile investment activities}\end{array}$$

Free cash flow extends cash flow from operating activities by recognizing that some of the cash a business generates must be plowed back into the business, in the form of capital expenditures, to support growth. Abstracting from a few technical details, free cash flow is essentially cash flow from operating activities less capital expenditures. As we will see in Chapter 9, free cash flow is a fundamental determinant of the value of a business. Indeed, one can argue that the principal means by which a company creates value for its owners is to increase free cash flow.

Yet another widely used cash flow measure is

$$\text{Discounted cash flow} = \begin{array}{l}\text{A sum of money today having the same value} \\ \text{as a future stream of cash receipts and disbursements}\end{array}$$

Discounted cash flow refers to a family of techniques for analyzing investment opportunities that take into account the time value of money. A standard approach to valuing investments and businesses uses discounted cash flow techniques to calculate the present value of projected free cash flows. This is the focus of the last three chapters of this book.

My advice when tossing cash flow terms about is to either use the phrase broadly to refer to a general movement of cash or to define your terms carefully.

and inventories, to support increasing sales. And although such investments reduce net cash provided by operating activities, they do not in any way suggest poor performance. Second, cash flow statements turn out to be less objective, and thus less immune to manipulation than might be supposed. Here's a

simple example. Suppose two companies are identical except that one sells its product on a simple open account, while the other loans its customers money enabling them to pay cash for the product. In both cases, the customer has the product and owes the seller money. But the increase in accounts receivable recorded by the first company on each sale will lower its cash flows from operating activities relative to the second, which can report the customer loan as part of investing activities. Because the criteria for apportioning cash flows among operating, investing, and financing activities are ambiguous, subjective judgment must be used in the preparation of cash flow statements.

Much of the information contained in a cash flow statement can be gleaned from careful study of a company's income statement and balance sheet. Nonetheless, the statement has three principal virtues. First, accounting neophytes and those who do not trust accrual accounting have at least some hope of understanding it. Second, the statement provides more accurate information about certain activities, such as share repurchases and employee stock options than one can infer from income statements and balance sheets alone. Third, it casts a welcome light on cash generation and solvency.

Financial Statements and the Value Problem

To this point, we have reviewed the basics of financial statements and grappled with the distinction between earnings and cash flow. This is a valuable start, but if we are to use financial statements to make informed business decisions, we must go further. We must understand the extent to which accounting numbers reflect economic reality. When the accountant tells us that Polaris's total assets were worth $4,922 million on December 31, 2020, is this literally true, or is the number just an artificial accounting construct? To gain perspective on this issue, and in anticipation of later discussions, I want to conclude by examining a recurring problem in the use of accounting information for financial decision making.

Market Value vs. Book Value

Part of what I will call the *value problem* involves the distinction between the market value and the book value of shareholders' equity. Polaris's 2020 balance sheet states that the value of shareholders' equity is $1,434 million. This is known as the *book value* of Polaris's equity. However, Polaris is not worth $1,434 million to its shareholders or to anyone else, for that matter. There are two reasons. One is that financial statements are largely *transactions-based.* If a company purchased an asset for $1 million in 1980, this transaction provides an objective measure of the asset's value, which the accountant uses to value the asset on the company's balance sheet. Unfortunately, it is a 1980 value that may or may not have much relevance today. To further

4. Explain briefly how each of the following transactions would affect a company's balance sheet. (Remember, assets must equal liabilities plus owners' equity before and after the transaction.)
   a. Sale of used equipment with a book value of $300,000 for $500,000 cash.
   b. Purchase of a new $80 million building, financed 40 percent with cash and 60 percent with a bank loan.
   c. Purchase of a new building for $60 million cash.
   d. A $40,000 payment to trade creditors.
   e. A firm's repurchase of 10,000 shares of its own stock at a price of $24 per share.
   f. Sale of merchandise for $80,000 in cash.
   g. Sale of merchandise for $120,000 on credit.
   h. A dividend payment to shareholders of $50,000.
5. Why do you suppose financial statements are constructed on an accrual basis rather than a cash basis when cash accounting is so much easier to understand?
6. Table 3.1 in Chapter 3 presents financial statements over the period 2018–2021 for R&E Supplies, Inc.
   a. Construct a sources and uses statement for the company over this period (one statement for all three years).
   b. What insights, if any, does the sources and uses statement give you about the financial position of R&E Supplies?
7. You are responsible for labor relations in your company. During heated labor negotiations, the General Secretary of your largest union exclaims, "Look, this company has $15 billion in assets, $7.5 billion in equity, and made a profit last year of $300 million—due largely, I might add, to the effort of union employees. So don't tell me you can't afford our wage demands." How would you reply?
8. You manage a real estate investment company. One year ago, the company purchased 10 parcels of land distributed throughout the community for $10 million each. A recent appraisal of the properties indicates that five of the parcels are now worth $8 million each, while the other five are worth $16 million each. Ignoring any income received from the properties and any taxes paid over the year, calculate the investment company's accounting earnings and its economic earnings in each of the following cases:
   a. The company sells all of the properties at their appraised values today.
   b. The company sells none of the properties.
   c. The company sells the properties that have fallen in value and keeps the others.

d. The company sells the properties that have risen in value and keeps the others.
e. After returning from a property management seminar, an employee recommends that the company adopt an end-of-year policy of always selling properties that have risen in value since purchase and always retaining properties that have fallen in value. The employee explains that, with this policy, the company will never show a loss on its real estate investment activities. Do you agree with the employee? Why, or why not?

9. During 2020, Acadia, Inc., earned net income of $500,000. The firm increased its accounts receivable during the year by $150,000. The book value of its assets declined by an amount equal to the year's depreciation charge, or $130,000, and the market value of its assets increased by $25,000. Based only on this information, how much cash did Acadia generate during the year? Please ignore taxes for this problem.
10. Desmond currently is a brewmaster for Copper Creek Brewery. He enjoys his job but is intrigued by the prospect of quitting and starting his own brewery. He currently makes $70,000 at Copper Creek. Desmond anticipates that his new brewery will have annual revenues of $250,000, and total annual expenses for operating the brewery, outside of any payments to Desmond, will be $190,000. Desmond comes to you with his idea. He believes that he would be equally happy with either option, but that starting his own brewery is the right decision in light of its profitability. Ignoring what might happen beyond the first year, do you agree with him? Why or why not?
11. Selected information for Blake's Restaurant Supply follows.

	($ millions)	
	2020	**2021**
Net sales	694	782
Cost of goods sold	450	502
Depreciation	51	61
Net income	130	142
Finished goods inventory	39	29
Accounts receivable	57	87
Accounts payable	39	44
Net fixed assets	404	482
Year-end cash balance	86	135

a. During 2021, how much cash did Blake's collect from sales?
b. During 2021, what was the cost of goods produced by the company?

c. Assuming the company neither sold nor salvaged any assets during the year, what were the company's capital expenditures during 2021?
d. Assuming there were no financing cash flows during 2021 and basing your answer solely on the information provided, what was Blake's cash flow from operations in 2021?

12. Below are summary cash flow statements for three roughly equal-sized companies.

	($ millions)		
	A	**B**	**C**
Net cash flows from operating activities	(300)	(300)	300
Net cash used in investing activities	(900)	(30)	(90)
Net cash from financing activities	1,200	210	(240)
Cash balance at beginning of the year	150	150	150

a. Calculate each company's cash balance at the end of the year.
b. Explain what might cause company C's net cash from financing activities to be negative.
c. Looking at companies A and B, which company would you prefer to own? Why?
d. Is company C's cash flow statement cause for any concern on the part of C's management or shareholders? Why or why not?

13. Coca-Cola recently had a market value of equity of $200 billion with 4 billion shares outstanding. The book value of its equity is $20 billion.
a. What is Coca-Cola's stock price per share? What is its book value per share?
b. If the company repurchases 100 million shares in the stock market at their current price, how will this affect the book value of equity if all else remains the same?
c. If there are no taxes or transaction costs, and investors do not change their perceptions of the firm, what should the market value of the firm be after the repurchase?
d. Instead of a share repurchase, the company decides to raise money by selling an additional 100 million shares on the market. If it can issue these additional shares at the current market price, how will this affect the book value of equity if all else remains the same?
e. If there are no taxes or transaction costs, and investors do not change their perception of the firm, what should the market value of the firm be after this stock issuance? What would its price per share be?

POLARIS

financial performance. Then, we broaden the discussion to consider the uses and limitations of ratio analysis as a tool for evaluating performance. To keep things practical, we will again use the financial statements for Polaris Inc., presented in Tables 1.2, 1.3, and 1.5 of the last chapter, to illustrate the techniques. The chapter concludes with an evaluation of Polaris's financial performance relative to its competition. At the end of the chapter, Table 2.4 presents summary definitions of the principal ratios appearing throughout the chapter.

Return on Equity

By far the most popular yardstick of financial performance among investors and senior managers is the *return on equity (ROE),* defined as

$$\text{Return on equity} = \frac{\text{Net income}}{\text{Shareholders' equity}}$$

Polaris's ROE for 2020 was

$$\text{ROE} = \frac{\$414}{\$1{,}434} = 28.9\%$$

It is not an exaggeration to say that the careers of many senior executives rise and fall with their firms' ROEs. ROE is accorded such importance because it is a measure of the *efficiency* with which a company employs owners' capital. It is a measure of earnings per dollar of invested equity capital or, equivalently, of the percentage return to owners on their investment. In short, it measures bang per buck.

Later in this chapter, we will consider some significant problems with ROE as a measure of financial performance. For now, let us accept it provisionally as at least widely used and see what we can learn.

The Three Determinants of ROE

To learn more about what management can do to increase ROE, suppose we rewrite ROE in terms of its three principal components:

$$\text{ROE} = \frac{\text{Net income}}{\text{Sales}} \times \frac{\text{Sales}}{\text{Assets}} \times \frac{\text{Assets}}{\text{Shareholders' equity}}$$

Denoting the last three ratios as the profit margin, asset turnover, and financial leverage, respectively, the expression can be written as

$$\text{Return on equity} = \text{Profit margin} \times \text{Asset turnover} \times \text{Financial leverage}$$

This says that management has only three levers for controlling ROE: (1) the earnings squeezed out of each dollar of sales, or the *profit margin;* (2) the sales

generated from each dollar of assets employed, or the *asset turnover;* and (3) the amount of equity used to finance the assets, or the *financial leverage.*[1] With few exceptions, whatever management does to increase these ratios increases ROE.

Note, too, the close correspondence between the levers of performance and company financial statements. Thus, the profit margin summarizes a company's income statement performance by showing profit per dollar of sales. The asset turnover ratio summarizes the company's management of the asset side of its balance sheet by showing the resources required to support sales. And the financial leverage ratio summarizes management of the liabilities side of the balance sheet by showing the amount of shareholders' equity used to finance the assets. This is reassuring evidence that, despite their simplicity, the three levers do capture the major elements of a company's financial performance.

We find that Polaris's 2020 ROE was generated as follows:

$$\frac{\$414}{\$1,434} = \frac{\$414}{\$7,108} \times \frac{\$7,108}{\$4,922} \times \frac{\$4,922}{\$1,434}$$

$$28.9\% = 5.824\% \times 1.444 \times 3.432$$

Table 2.1 presents ROE and its three principal components for 10 highly diverse businesses. It shows quite clearly that there are many paths to heaven: Even for companies with ROEs that are quite similar, the combinations of profit margin, asset turnover, and financial leverage producing these end results vary widely. ROE ranges from a high of 37.4 percent for Microsoft to a low of 6.1 percent for financial services company Citigroup, while the range for the profit margin, to take one example, is from a low of 2.0 percent for grocery retailer Kroger to a high of 31.0 percent for Microsoft. ROE differs by about 6 to 1 high to low, but the profit margin varies by a factor of almost 16 to 1. Comparable ranges for asset turnover and financial leverage are 68 to 1 and 9 to 1, respectively.

Why are ROEs similar across firms, while profit margins, asset turnovers, and financial leverages differ dramatically? The answer, in a word, is competition. Attainment of an unusually high ROE by one company acts as a magnet to attract rivals anxious to emulate the superior performance. As rivals enter the market, the heightened competition drives the successful company's ROE back toward the average. Conversely, unusually low ROEs repel potential new competitors and drive existing companies out of business so that, over time, survivors' ROEs rise toward the average.

To understand how managerial decisions and a company's competitive environment combine to affect ROE, we will examine each lever of

[1]At first glance the ratio of assets to shareholders' equity may not look like a measure of financial leverage, but consider the following

$$\frac{\text{Assets}}{\text{Equity}} = \frac{\text{Liabilities + Equity}}{\text{Equity}} = \frac{\text{Liabilities}}{\text{Equity}} + 1$$

And the liabilities-to-equity ratio clearly measures financial leverage.

TABLE 2.1 **ROEs and Levers of Performance for 10 Diverse Companies, 2020***

	Return on Equity (ROE)		Profit Margin (P)		Asset Turnover (A)		Financial Leverage (T)
	(%)	=	(%)	×	(times)	×	(times)
Alphabet Inc.	18.1	=	22.1	×	0.57	×	1.44
Citigroup	6.1	=	12.4	×	0.04	×	12.56
Diageo PLC	20.8	=	12.0	×	0.35	×	4.92
Kroger Co.	27.0	=	2.0	×	2.72	×	5.08
Microsoft Corp.	37.4	=	31.0	×	0.47	×	2.55
Norfolk Southern	13.6	=	20.6	×	0.26	×	2.57
Novartis AG	14.3	=	16.6	×	0.37	×	2.33
Polaris Inc.	28.9	=	5.8	×	1.44	×	3.43
Target Corp.	30.2	=	4.7	×	1.83	×	3.55
Wendy's Co.	21.4	=	6.8	×	0.34	×	9.17

*Totals may not add due to rounding.

performance in more detail. In anticipation of the discussion of ratio analysis to follow, we will also consider related commonly used financial ratios.

The Profit Margin

The profit margin measures the fraction of each dollar of sales that trickles down through the income statement to profits. This ratio is particularly important to operating managers because it reflects the company's pricing strategy and its ability to control operating costs. As Table 2.1 indicates, profit margins differ greatly among industries depending on the nature of the product sold and the company's competitive strategy.

Note too that profit margin and asset turnover tend to vary inversely. Companies with high profit margins tend to have low asset turns, and vice versa. This is no accident. Companies that add significant value to a product, such as Microsoft or Alphabet Inc., can demand high profit margins. However, because adding value to a product usually requires lots of assets, these same firms tend to have lower asset turns. At the other extreme, grocery stores such as Kroger and discount retailers such as Target bring the product in the store on forklift trucks, sell for cash, and make the customer carry out her own purchases. Because they add little value to the product, they have very low profit margins and correspondingly high asset turns. It should be apparent, therefore, that a high profit margin is not necessarily better or worse than a low one—it all depends on the combined effect of the profit margin and the asset turnover.

Return on Assets

To look at the combined effect of margins and turns, we can calculate the *return on assets (ROA):*

$$\text{ROA} = \underset{\text{margin}}{\text{Profit}} \times \underset{\text{turnover}}{\text{Asset}} = \frac{\text{Net income}}{\text{Assets}}$$

Polaris's ROA in 2020 was

$$\text{Return on assets} = \frac{\$414}{\$4,922} = 8.4\%$$

This means Polaris earned an average of 8.4 cents on each dollar tied up in the business.

ROA is a basic measure of the efficiency with which a company allocates and manages its resources. It differs from ROE, in that it measures profit as a percentage of the money provided by owners *and* creditors as opposed to only the money provided by owners.

Some companies, such as Alphabet Inc. and Novartis, produce their ROAs by combining a high profit margin with a low asset turn; others, such as Target and Kroger, adopt the reverse strategy. A high profit margin *and* a high asset turn are ideal, but can be expected to attract considerable competition. Conversely, a low profit margin combined with a low asset turn will attract only bankruptcy lawyers.

Gross Margin

When analyzing profitability, it is often interesting to distinguish between variable costs and fixed costs. Variable costs change as sales vary, while fixed costs remain constant. Companies with a high proportion of fixed costs are more vulnerable to sales declines than other firms, because they cannot reduce fixed costs as sales fall.

Unfortunately, the accountant does not differentiate between fixed and variable costs when constructing an income statement. However, it is usually safe to assume that most expenses in cost of goods sold are variable, while most of the other operating costs are fixed. The gross margin enables us to distinguish, insofar as possible, between fixed and variable costs. It is defined as

$$\text{Gross margin} = \frac{\text{Gross profit}}{\text{Sales}} = \frac{\$2,051}{\$7,108} = 28.9\%$$

where gross profit equals net sales less cost of goods sold. Twenty-nine percent of Polaris's sales dollar is a *contribution to fixed cost and profits:* 29 cents of every sales dollar is available to pay for fixed costs and to add to profits.

One common use of the gross margin is to estimate a company's breakeven sales volume. Polaris's income statement tells us that total operating expenses

in 2020 were $1,460 million. If we assume these expenses are fixed and if 29 cents of each Polaris sales dollar is available to pay for fixed costs and add to profits, the company's zero-profit sales volume must be $1,460/0.29, or $5,034 million.[2] Assuming operating expenses and the gross margin are independent of sales, Polaris loses money when sales are below $5,034 million, and makes money when sales are above this figure.

Asset Turnover

Some newcomers to finance believe assets are a good thing: the more the better. The reality is just the opposite: Unless a company is about to go out of business, its value is in the income stream it generates, and its assets are simply a necessary means to this end. Indeed, the ideal company would be one that produced income without any assets; then no investment would be required, and returns would be infinite. Short of this fantasy, our ROE equation tells us that, other things constant, financial performance improves as asset turnover rises. This is the second lever of management performance.

The asset turnover ratio measures the sales generated per dollar of assets. Polaris's asset turnover of 1.44 means that Polaris generated $1.44 of sales for each dollar invested in assets. This ratio measures asset intensity, with a low asset turnover signifying an asset-intensive business and a high turnover the reverse.

The nature of a company's products and its competitive strategy strongly influence asset turnover. A steel mill will never have the asset turnover of a grocery store. But this is not the end of the story, because management diligence and creativity in controlling assets are also vital determinants of a company's asset turnover. When product technology is similar among competitors, control of assets is often the margin between success and failure.

Control of current assets is especially critical. You might think the distinction between current and fixed assets based solely on whether the asset will revert to cash within one year is artificial. But more is involved than this. Current assets, especially accounts receivable and inventory, have several unique properties. One is that if something goes wrong—if sales decline unexpectedly, customers delay payment, or a critical part fails to arrive—a company's investment in current assets can balloon very rapidly. When even manufacturing companies routinely invest one-half or more of their money in current assets, it is easy to appreciate that even modest alterations in the management of these assets can significantly affect company finances.

A second distinction is that unlike fixed assets, current assets can become a source of cash during business downturns. As sales decline, a company's investment in accounts receivable and inventory should fall as well, thereby freeing cash for other uses. (Remember, a reduction in an asset account is a

[2]Income = Sales − Variable costs − Fixed costs = Sales × Gross margin − Fixed costs. Setting income to zero and solving for sales, Sales = Fixed costs/Gross margin.

source of cash.) The fact that in a well-run company current assets move in an accordion-like fashion with sales is appealing to creditors. They know that during the upswing of a business cycle rising current assets will require loans, while during a downswing, falling current assets will provide the cash to repay the loans. In bankers' jargon, such a loan is said to be *self-liquidating* in the sense that the use to which the money is put creates the source of repayment.

It is often useful to analyze the turnover of each type of asset on a company's balance sheet individually. This gives rise to what are known as *control ratios.* Although the form in which each ratio is expressed may vary, every control ratio is simply an asset turnover for a particular type of asset. In each instance, the firm's investment in the asset is compared to net sales or a closely related figure.

Why compare assets to sales? The fact that a company's investment in, say, accounts receivable has risen over time could be due to two forces: (1) Perhaps sales have risen and simply dragged receivables along, or (2) management may have slackened its collection efforts. Relating receivables to sales in a control ratio adjusts for changes in sales, enabling the analyst to concentrate on the more important effects of changing management control. Thus, the control ratio distinguishes between sales-induced changes in investment and other, perhaps more sinister causes. Following are some standard control ratios and their values for Polaris in 2020.

Inventory Turnover

Inventory turnover is expressed as

$$\text{Inventory turnover} = \frac{\text{Cost of goods sold}}{\text{Ending inventory}} = \frac{\$5,057}{\$1,178} = 4.3 \text{ times}$$

An inventory turn of 4.3 times means that items in Polaris's inventory turn over 4.3 times per year, on average. Said differently, the typical item sits in inventory about 85 days before being sold (365 days/4.3 times = 85 days). This last number is known as *days inventory outstanding* (or *days' sales in inventory*), because Polaris has enough inventory to support 85 days' worth of sales.

Several alternative definitions of the inventory turnover ratio exist, including sales divided by ending inventory and cost of goods sold divided by average inventory. Cost of goods sold is a more appropriate numerator than sales because sales include a profit markup that is absent from inventory. But beyond this, I see little to choose from among the various definitions.

The Collection Period

The *collection period* highlights a company's management of accounts receivable. For Polaris

$$\text{Collection period} = \frac{\text{Accounts receivable}}{\text{Credit sales per day}} = \frac{\$257}{\$7,108/365} = 13.2 \text{ days}$$

Credit sales appear here rather than net sales because only credit sales generate accounts receivable. As a company outsider, however, I do not know what portion of Polaris's net sales, if any, are for cash, so I assume they are all on credit. Credit sales per day is defined as credit sales for the accounting period divided by the number of days in the accounting period, which for annual statements is obviously 365 days.

Two interpretations of Polaris's collection period are possible. We can say that Polaris has an average of 13 days' worth of credit sales tied up in accounts receivable, or we can say that the average time lag between sale and receipt of cash from the sale is 13 days.

Beware of Seasonal Companies

Interpreting ratios of companies with seasonal sales can be tricky. For example, suppose a company's sales peak sharply at Christmas, resulting in high year-end accounts receivable. A naive collection period calculated by relating year-end accounts receivable to average daily sales for the whole year will produce an apparently very high collection period because the denominator is insensitive to the seasonal peak. To avoid being misled, a better way to calculate the collection period for a seasonal company is to use credit sales per day based only on the prior 60 to 90 days' sales. This matches the accounts receivable to the credit sales actually generating the receivables.

If we like, we can define a simpler asset turnover ratio for accounts receivable as just credit sales/accounts receivable. However, the collection period format is more informative, because it allows us to compare a company's collection period with its terms of sale. Thus, if a company sells on 90-day terms, a collection period of 65 days is excellent, but if the terms of sale were 30 days, our interpretation would be quite different.

Days' Sales in Cash

Polaris's *days' sales in cash* is

$$\text{Days' sales in cash} = \frac{\text{Cash and securities}}{\text{Sales per day}} = \frac{\$544}{\$7,108/365} = 28.0 \text{ days}$$

Polaris has 28 days' worth of sales in cash and securities. Whether this is the right amount of cash for Polaris is difficult to say. On the one hand, cash balances should not be too low. Companies require modest amounts of cash to facilitate transactions and are sometimes required to carry substantially larger amounts as compensating balances for bank loans. In addition, cash and marketable securities can be an important source of liquidity for a firm in an emergency. On the other hand, if cash balances are too high, shareholders may become disappointed that the firm's assets are not put to more productive and profitable uses. So the question of how much cash and securities a

company should carry is often closely related to the broader question of how important liquidity is to the company and how best to provide it. The median days' sales in cash for nonfinancial companies in the Standard & Poor's 500 Index in 2020 was 102 days. In comparison, Polaris's 28 days is low. Relative to information technology companies, which often carry notoriously high cash balances, Polaris's cash looks even more modest. The median days' sales in cash in 2020 among the information technology companies in the S&P 500 was 155 days, with Microsoft and Oracle clocking in at 348 and 402 days, respectively.

Alphabet's Levers of Performance

The 2020 levers of performance for internet titan Alphabet (the parent company of Google) make instructive reading. As shown below, the company combined an attractive profit margin and conservative financial leverage with an asset turnover of only 0.57 times to generate a rather ordinary ROE of 18.1 percent. This is mediocre performance for a company selling at over 29 times earnings and perceived by most to be the dominant internet player.

How can an internet company generate an asset turnover more like that of a steel mill or a public utility? A look at Alphabet's balance sheet explains the mystery. At year-end 2020, roughly $137 billion, or over 40 percent of Alphabet's assets, were in cash and marketable securities, and its days' sales in cash was a whopping 273. It's as if the company had merged with a money market mutual fund. And Alphabet is not alone. It is common practice among leading technology companies to build huge war chests, which they argue are necessary to finance continued growth and to facilitate possible acquisitions—like maybe if Panama or South Dakota ever came up for sale. Others, including Ralph Nader, see a more sinister purpose: to keep the money out of the hands of shareholders and to avoid taxes.

To focus on Alphabet's operating performance apart from its ability to invest excess cash, we can strip cash and marketable securities out of the analysis. To do this, imagine the company returned 90 percent of its cash and securities to shareholders as a giant dividend. Alternatively, imagine Alphabet split into two companies: an operating internet company and a money market mutual fund charged with investing 90 percent of the firm's cash. This would cut the operating company's assets and shareholders' equity by $123 billion, while leaving the company with a still robust 27 days' sales in cash. Assuming a modest 1 percent after-tax return on cash and securities, this would knock $1.23 billion from net income. The resulting revised levers of performance appear in the following summary. Asset turnover is now a more plausible, but still modest, 0.93 times, and ROE is up to a robust 39.4 percent. These numbers more accurately reflect the economics of Alphabet's business.

	Return on Equity	=	Profit Margin	×	Asset Turnover	×	Financial Leverage
As reported	18.1%	=	22.1%	×	0.57	×	1.44
Revised	39.4%	=	21.4%	×	0.93	×	1.98

Payables Period

The *payables period* is a control ratio for a liability. It is simply the collection period applied to accounts payable. For Polaris

$$\text{Payables period} = \frac{\text{Accounts payable}}{\text{Credit purchases per day}} = \frac{\$782}{\$5{,}057/365} = 56.4 \text{ days}$$

The proper definition of the payables period uses credit purchases, because they are what generate accounts payable. However, an outsider seldom knows credit purchases, so it is frequently necessary to settle for the closest approximation: cost of goods sold. This is what I have done above for Polaris; $5,057 million is Polaris's cost of goods sold, not its credit purchases. Cost of goods sold can differ from credit purchases for two reasons. First, the company may be adding to or depleting inventory—that is, purchasing at a different rate than it is selling. Second, all manufacturers add labor to material in the production process, thereby making cost of goods sold larger than purchases. Because of these differences, it is tricky to compare a manufacturing company's payables period, based on cost of goods sold, to its purchase terms. For Polaris, it is almost certain that cost of goods sold overstates credit purchases per day and that Polaris's suppliers are waiting a good bit longer than 56.4 days, on average, to receive payment.

Fixed-Asset Turnover

Companies or industries requiring large investments in long-lived assets to produce their goods are said to be capital intensive. Because a preponderance of their costs are fixed, capital-intensive businesses, such as auto manufacturers and airlines, are especially sensitive to the state of the economy, prospering in good times as sales rise relative to costs and suffering in bad as the reverse occurs. Capital intensity, also referred to as operating leverage, is of particular concern to creditors because it magnifies the basic business risks faced by a firm.

Fixed-asset turnover is a measure of capital intensity, with a low turnover implying high intensity. The ratio in 2020 for Polaris is

$$\begin{aligned}\text{Fixed-asset turnover} &= \frac{\text{Sales}}{\text{Net property, plant, and equipment}} = \frac{\$7,108}{\$1,014} \\ &= 7.0 \text{ times}\end{aligned}$$

where $1,014 million is the book value of Polaris's net property, plant, and equipment.

Operating and Cash Conversion Cycles

Several of these working capital control ratios are related, and it often makes sense for managers to recognize the connections. When a company begins its production cycle, it uses cash to acquire inventory. When the company sells an item, inventory changes back into cash once the customer pays for it and the accounts receivable is collected. In Chapter 1, we spoke of this movement of cash in qualitative terms as the firm's operating, or working capital, cycle. Here, we can express the same notion numerically as the sum of the number of days cash is tied up in inventory plus the number of days it takes a customer to pay, or the collection period.

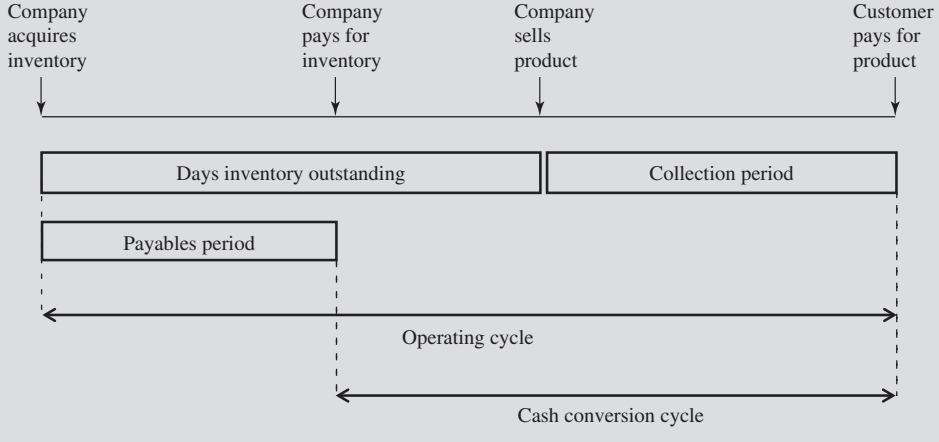
$$\text{Operating cycle} = \text{Days inventory outstanding} + \text{Collection period}$$

For Polaris, days inventory outstanding is 85 days and the collection period is 13 days, so the total operating cycle is the sum: 98 days. Polaris's cash is tied up for an average of 98 days between when it acquires inventory and when its customers pay. So Polaris must find a source for this cash.

The good news for Polaris is that suppliers are willing to provide some of this cash by allowing Polaris some time to pay suppliers for the inventory it acquired. Recognizing this, the cash conversion cycle starts with the operating cycle and nets out the length of time Polaris uses its suppliers' money:

$$\begin{aligned}\text{Cash conversion cycle (CCC)} &= \text{Operating cycle} - \text{Payables period}\\ &= \text{Days inventory outstanding}\\ &\quad + \text{Collection period} - \text{Payables period}\end{aligned}$$

Here is a timeline for the operating and cash conversion cycles.



Polaris's cash conversion cycle is its operating cycle of 98 days minus its payables period of 56 days, for a net of 42 days. Although Polaris must begin the process by acquiring inventory, it is able to wait 56 days, on average, before paying for this inventory. So the length of time between when cash goes out the door until it comes back in, the cash conversion cycle, is 42 days.

Financial Leverage

The third lever by which management affects ROE is financial leverage. A company increases its financial leverage when it raises the proportion of debt relative to equity used to finance the business. Unlike the profit margin and the asset turnover ratio, where more is generally preferred to less, financial leverage is not something management necessarily wants to maximize, even when doing so increases ROE. Instead, the challenge of financial leverage is to strike a prudent balance between the benefits and costs of debt financing. Later, we will devote all of Chapter 6 to this important financial decision. For now, it is sufficient to recognize that more leverage is not necessarily preferred to less, and that while companies have considerable latitude in their choice of how much financial leverage to employ, there are economic and institutional constraints on their decision.

As Table 2.1 suggests, the nature of a company's business and its assets influence the financial leverage it can employ. In general, businesses with highly predictable and stable operating cash flows, such as Diageo, an alcoholic beverage company, can safely undertake more financial leverage than firms facing a high degree of market uncertainty, such as pharmaceutical company Novartis. In addition, businesses such as banks, which usually have diversified portfolios of readily salable, liquid assets, can also safely use more financial leverage than the typical business.

Another pattern evident in Table 2.1 is that ROA and financial leverage tend to be inversely related. Companies with low ROAs generally employ more debt financing, and vice versa. This is consistent with the previous paragraph. Safe, stable, liquid investments tend to generate low returns but substantial borrowing capacity. Banks are extreme examples of this pattern. Citigroup combines what by manufacturing standards would be a horrible 0.5 percent ROA with an astronomical leverage ratio of 12.56 to generate a modest ROE of 6.1 percent. The key to this pairing is the safe, liquid nature of the bank's assets. (Past loans to Third-World dictators, Texas energy companies, and subprime mortgage borrowers are, of course, another story—one the banks would just as soon forget.)

The following ratios measure financial leverage, or debt capacity, and the related concept of liquidity.

Balance Sheet Ratios

The most common measures of financial leverage compare the book value of a company's liabilities to the book value of its assets or equity. This gives rise to the *debt-to-assets ratio* and the *debt-to-equity ratio,* defined as

$$\text{Debt-to-assets ratio} = \frac{\text{Total liabilities}}{\text{Total assets}} = \frac{\$3,488}{\$4,922} = 70.9\%$$

$$\text{Debt-to-equity ratio} = \frac{\text{Total liabilities}}{\text{Shareholders' equity}} = \frac{\$3,488}{\$1,434} = 243.3\%$$

The first ratio says that money to pay for 71 percent of Polaris's assets, in book value terms, comes from creditors of one type or another. The second ratio says the same thing in a slightly different way: Creditors supply Polaris with $2.43 for every dollar supplied by shareholders. As footnote 1 demonstrated, the lever of performance introduced earlier, the assets-to-equity ratio, is just the debt-to-equity ratio plus 1.

As many companies have built up large excess balances of cash and marketable securities, analysts have increasingly replaced debt in these equations with "net" debt, defined as total liabilities less cash and marketable securities. The idea is that as safe, interest-bearing assets, excess cash and securities are essentially negative debt and should thus be subtracted from liabilities when measuring aggregate indebtedness. Thought of another way, if a company has excess cash that could, on short notice, be used to pay off outstanding debts, then the company's true level of indebtedness is lower than the standard leverage ratios would suggest. Because, as noted in Chapter 1, Polaris's cash balance increased markedly in 2020, calculating their debt ratios using net debt could be enlightening. Indeed, the net debt figures paint a rather different picture of Polaris's indebtedness, with a debt-to-assets ratio of only 59.8 percent and a debt-to-equity ratio of 205.3 percent.

Coverage Ratios

A number of variations on these balance sheet measures of financial leverage exist. Conceptually, however, there is no reason to prefer one over another, for they all focus on balance sheet values, and hence all suffer from the same weakness. The financial burden a company faces by using debt financing ultimately depends not on the size of its liabilities relative to assets or to equity but on its ability to meet the annual cash payments the debt requires. A simple example will illustrate the distinction. Suppose two companies, A and B, have the same debt-to-assets ratio, but A is very profitable and B is losing money. Chances are that B will have difficulty meeting its annual interest and principal obligations, while A will not. The obvious conclusion is that balance sheet ratios are of primary interest only in liquidation, when the proceeds of asset sales are to be distributed among creditors and owners. In all other instances, we should be more interested in comparing the annual burden the debt imposes to the cash flow available for debt service.

This gives rise to what are known as *coverage ratios,* the most common of which are *times interest earned* and *times burden covered.* Letting EBIT represent earnings before interest and taxes (calculated as earnings before taxes plus interest expense), the ratios are defined as

$$\text{Times interest earned} = \frac{\text{EBIT}}{\text{Interest expense}} = \frac{\$588}{\$67} = 8.8 \text{ times}$$

$$\text{Times burden covered} = \frac{\text{EBIT}}{\text{Interest} + \dfrac{\text{Principal repayment}}{1 - \text{Tax rate}}}$$

$$= \frac{\$588}{\$67 + \dfrac{\$142}{1 - \left(\dfrac{\$107}{\$521}\right)}} = 2.4 \text{ times}$$

Both ratios compare income available for debt service in the numerator to some measure of annual financial obligation. For both ratios, the income available is EBIT. This is the earnings the company generates that can be used to make interest payments. EBIT is before taxes because interest payments are before-tax expenditures, and we want to compare like quantities. Polaris's times interest earned ratio of 8.8 means the company earned its interest obligation 8.8 times over in 2020; EBIT was 8.8 times as large as interest.

Though dentists may correctly claim that if you ignore your teeth they'll eventually go away, the same cannot be said for principal repayments. If a company fails to make a principal repayment when due, the outcome is the same as if it had failed to make an interest payment. In both cases, the company is in default and creditors can force it into bankruptcy. The times-burden-covered ratio reflects this reality by expanding the definition of annual financial obligations to include debt principal repayments, as well as interest. When including principal repayment as part of a company's financial burden, we must remember to express the figure on a before-tax basis comparable to interest and EBIT. Unlike interest payments, principal repayments are not a tax-deductible expense. This means that if a company is in, say, a 50 percent tax bracket, it must earn $2 before taxes to have $1 after taxes to pay creditors. The other dollar goes to the tax collector. For other tax brackets, the before-tax burden of a principal repayment is found by dividing the repayment by 1 minus the company's tax rate. Adjusting the principal repayment in this manner to its before-tax equivalent is known in the trade as *grossing up* the principal—about as gross as finance ever gets.

An often-asked question is: Which of these coverage ratios is more meaningful? The answer is that both are important. If a company could always roll over its maturing obligations by taking out new loans as it repaid old ones, the *net* burden of the debt would be merely the interest expense, and times interest earned would be the more important ratio. The problem, as we were all vividly reminded during the recent financial panic, is that the replacement of maturing debt with new debt is not an automatic feature of capital markets.

In some instances, when capital markets are unsettled or a company's fortunes decline, creditors may refuse to renew maturing obligations. Then, the burden of the debt suddenly becomes interest plus principal payments, and the times-burden-covered ratio assumes paramount importance.

This is what happened beginning in the summer of 2007 when growing defaults on subprime mortgages prompted some short-term lenders to demand immediate payment from a variety of mortgage investment companies. These special purpose companies were issuing short-term debt to finance ownership of complex, long-term mortgage-backed securities. This was a nice business as long as lenders willingly rolled over maturing debts. But the minute they balked, a vicious circle ensued as borrowers sold their securities at cut-rate prices to repay short-term lenders, and short-term lenders, reacting to the falling prices, increasingly refused to roll over maturing obligations.

In sum, it is fair to conclude that the times-burden-covered ratio is too conservative because it assumes the company will pay its existing loans down to zero, but that the times-interest-earned ratio is too liberal because it assumes the company will always roll over all of its obligations as they mature.

Market Value Leverage Ratios

A third family of leverage ratios relates a company's liabilities to the *market value of its assets* or the *market value of its equity.* For Polaris in 2020,

$$\frac{\text{Market value of debt}}{\text{Market value of assets}} = \frac{\text{Market value of debt}}{\text{Market value of debt } + \text{ equity}}$$

$$= \frac{\$3,488}{\$3,488 + \$5,898} = 37.2\%$$

$$\frac{\text{Market value of debt}}{\text{Market value of equity}} = \frac{\text{Market value of debt}}{\text{Number of shares of stock} \times \text{Price per share}}$$

$$= \frac{\$3,488}{\$5,898} = 59.1\%$$

Note that I have assumed the market value of debt equals the book value of debt in both of these ratios. Strictly speaking, this is seldom true, but in most instances, the difference between the two quantities is small. Also, accurately estimating the market value of debt often turns out to be a tedious, time-consuming chore that is best avoided—unless, of course, you are being paid by the hour.

Market value ratios are clearly superior to book value ratios simply because book values are historical, often irrelevant numbers, while market values indicate the true worth of creditors' and owners' stakes in the business. Recalling that market values are based on investors' expectations about future cash flows, market value leverage ratios can be thought of as coverage ratios extended over

many future periods. Instead of comparing income to financial burden in a single year as coverage ratios do, market value ratios compare today's value of expected future income to today's value of future financial burdens.

Market value ratios are especially helpful when assessing the financial leverage of rapidly growing, startup businesses. Even when such companies have terrible or nonexistent coverage ratios, lenders may still extend them liberal credit if they believe future cash flows will be sufficient to service the debt. McCaw Communications offers an extreme example of this. At year-end 1990, McCaw had over $5 billion in debt; a debt-to-equity ratio, in book terms, of 330 percent; and annualized interest expenses of *more than 60 percent of net revenues.* Moreover, despite explosive growth, McCaw had never made a meaningful operating profit in its principal cellular telephone business. Why then did otherwise intelligent creditors loan McCaw $5 billion? Because creditors and equity investors believed it was only a matter of time before the company would begin to generate huge cash flows. This optimism was handsomely rewarded in late 1993 when AT&T paid $12.6 billion to acquire McCaw. Including the $5 billion in debt assumed by AT&T, the acquisition ranked as the second largest in corporate history at the time.

Another example is Amazon. In 1998, the company recorded its largest-ever loss of $124 million, had never earned a profit, and had only $139 million left in shareholders' equity. But not to worry: Lenders were still pleased to extend the company $350 million in long-term debt. Apparently, creditors are willing to overlook a number of messy details when a borrower's sales are growing 300 percent a year and the market value of its equity tops $17 billion—especially when the debt is convertible into equity. After all, in market value terms, Amazon's debt-to-equity ratio was only 3 percent. Today, Amazon's equity is worth $1.75 trillion, and its market-value debt-to-equity ratio is about 13 percent.

Economists like market value leverage ratios because they are accurate indicators of company indebtedness at a point in time. But you should be aware that market value ratios are not without problems. One is that they ignore rollover risks. When creditors take the attitude that debt must be repaid with cash, not promises of future cash, modest market value leverage ratios can be of hollow comfort. Also, despite these ratios' conceptual appeal, few companies use them to set financing policy or to monitor debt levels. This may be due in part to the fact that volatile stock prices can make market value ratios appear somewhat arbitrary and beyond management's control.

Liquidity Ratios

One determinant of a company's debt capacity is the liquidity of its assets. An asset is liquid if it can be readily converted to cash, while a liability is liquid if it must be repaid in the near future. As the subprime mortgage debacle

illustrates, it is risky to finance illiquid assets such as fixed plant and equipment with liquid, short-term liabilities, because the liabilities will come due before the assets generate enough cash to pay them. Such "maturity mismatching" forces borrowers to roll over, or refinance, maturing liabilities to avoid insolvency.

Two common ratios intended to measure the liquidity of a company's assets relative to its liabilities are the *current ratio* and the *acid test.* For Polaris,

$$\text{Current ratio} = \frac{\text{Current assets}}{\text{Current liabilities}}$$
$$= \frac{\$2,117}{\$1,889} = 1.1 \text{ times}$$
$$\text{Acid test} = \frac{\text{Current assets} - \text{Inventory}}{\text{Current liabilities}}$$
$$= \frac{\$2,117 - \$1,178}{\$1,889} = 0.5 \text{ times}$$

The current ratio compares the assets that will turn into cash within the year to the liabilities that must be paid within the year. A company with a low current ratio lacks liquidity, in the sense that it cannot reduce its current assets for cash to meet maturing obligations. It must rely instead on operating income and outside financing.

The acid-test ratio, sometimes called the *quick ratio,* is a more conservative liquidity measure. It is identical to the current ratio except that the numerator is reduced by the value of inventory. Inventory is subtracted because it is frequently illiquid. Under distress conditions, a company or its creditors may realize little cash from the sale of inventory. In liquidation sales, sellers typically receive 40 percent or less of the book value of inventory.

You should recognize that these ratios are rather crude measures of liquidity, for at least two reasons. First, rolling over some obligations, such as accounts payable, involves virtually no insolvency risk, provided the company is at least marginally profitable. Second, unless a company intends to go out of business, most of the cash generated by liquidating current assets cannot be used to reduce liabilities because it must be plowed back into the business to support continued operations.

Is ROE a Reliable Financial Yardstick?

To this point, we have assumed management wants to increase the company's ROE, and we have studied three important levers of performance by which they can accomplish this: the profit margin, asset turnover, and

financial leverage. We concluded that whether a company is IBM or the corner drugstore, careful management of these levers can positively affect ROE. We also saw that determining and maintaining appropriate values of the levers is a challenging managerial task that requires an understanding of the company's business, the way the company competes, and the interdependencies among the levers themselves. Now it is time to ask how reliable ROE is as a measure of financial performance. If company A has a higher ROE than company B, is it necessarily a better company? If company C increases its ROE, is this unequivocal evidence of improved performance?

ROE suffers from three critical deficiencies as a measure of financial performance, which I will refer to as the *timing* problem, the *risk* problem, and the *value* problem. Seen in proper perspective, these problems mean ROE is seldom an unambiguous measure of performance. ROE remains a useful and important indicator, but it must be interpreted in light of its limitations, and no one should automatically assume a higher ROE is always better than a lower one.

The Timing Problem

It is a cliché to say that successful managers must be forward looking and have a long-term perspective. Yet ROE is precisely the opposite: backward looking and focused on a single year. So it is little wonder that ROE can at times be a skewed measure of performance. When, for example, a company incurs heavy startup costs to introduce a hot new product, ROE will initially fall. However, rather than indicating worsening financial performance, the fall simply reflects the myopic, one-period nature of the yardstick. Because ROE necessarily includes only one year's earnings, it fails to capture the full impact of multiperiod decisions.

The Risk Problem

Business decisions commonly involve the classic "eat well–sleep well" dilemma. If you want to eat well, you had best be prepared to take risks in search of higher returns. If you want to sleep well, you will likely have to forgo high returns in search of safety. Seldom will you realize both high returns and safety. (And when you do, please give me a call.)

The problem with ROE is that it says nothing about what risks a company has taken to generate it. Here is a simple example. Take-a-Risk, Inc., earns an ROA of 6 percent from wildcat oil exploration in Sudan, which it combines with an assets-to-equity ratio of 5.0 to produce an ROE of 30 percent ($6\% \times 5.0$). Never-Dare, Ltd., meanwhile, has an ROA of 10 percent on its investment in government securities, which it finances with equal portions of debt and equity, yielding an ROE of 20 percent ($10\% \times 2.0$). Which company is the

better performer? My answer is Never-Dare. Take-a-Risk's ROE is high, but its high business risk and extreme financial leverage make it a very uncertain enterprise. I would prefer the more modest but eminently safer ROE of Never-Dare.[3] Security analysts would make the same point by saying that Take-a-Risk's ROE might be higher, but that the number is much lower in quality than Never-Dare's ROE, meaning that it is much riskier. In sum, because ROE looks only at return while ignoring risk, it can be an inaccurate yardstick of financial performance.

Return on Invested Capital

To circumvent the distorting effects of leverage on ROE and ROA, I recommend calculating *return on invested capital (ROIC),* also known as *return on net assets (RONA):*

$$\text{ROIC} = \frac{\text{EBIT}(1 - \text{Tax rate})}{\text{Interest-bearing debt} + \text{Equity}}$$

Polaris's 2020 ROIC was

$$\frac{\$588(1 - \$107/\$521)}{\$142 + \$1,294 + \$1,434} = 16.3\%$$

The numerator of this ratio is the earnings after tax the company would report if it were all equity financed, and the denominator is the sum of all sources of cash to the company on which a return must be earned. Thus, while accounts payable are a source of cash to the company, they are excluded because they carry no explicit cost. In essence, ROIC is the rate of return earned on the total capital invested in the business without regard for whether it is called debt or equity.[4]

To see the virtue of ROIC, consider the following example. Companies A and B are identical in all respects except that A is highly levered and B is all equity financed. Because the two companies are identical except for capital structure, we would like a return measure that reflects this fundamental similarity. The following table shows that ROE and ROA fail this test. Reflecting the company's extensive use of financial leverage, A's ROE is 22.5 percent, while B's zero-leverage position generates a lower but better-quality ROE of

[3]Even if I preferred eating well to sleeping well, I would still choose Never-Dare and finance my purchase with a little personal borrowing to lever my return on investment. See the appendix to Chapter 6 for more on the substitution of personal borrowing for company borrowing.

[4]After decades of neglect, companies appear to be rediscovering ROIC. For more, see David Benoit, "The Hottest Metric in Finance: ROIC," *The Wall Street Journal*, May 3, 2016.

9.0 percent. ROA is biased in the other direction, punishing company A for its extensive use of debt and leaving B unaffected. Only ROIC is independent of the different financing schemes the two companies employ, showing a 9 percent return for both firms. ROIC thus reflects the company's fundamental earning power before it is confounded by differences in financing strategies.

	Company	
	A	**B**
Debt @ 10% interest	$ 900	$ 0
Equity	100	1,000
Total assets	$1,000	$1,000
EBIT	$ 120	$ 120
– Interest expense	90	0
Earnings before tax	30	120
– Tax @ 25%	8	30
Earnings after tax	$ 23	$ 90
ROE	22.5%	9.0%
ROA	2.3%	9.0%
ROIC	9.0%	9.0%

The Value Problem

ROE measures the return on shareholders' investment; however, the investment figure used is the *book value* of shareholders' equity, not the *market value.* This distinction is important. Polaris's ROE in 2020 was 28.9 percent, and indeed this is the return you could have earned had you been able to buy the company's equity for its book value of $1,434 million. But that would have been impossible, for, as noted in the previous chapter, the market value of Polaris's equity was $5,898 million. At this price, your annual return would have been only 7.0 percent, not 28.9 percent ($414/$5,898 = 7.0%). The market value of equity is more significant to shareholders because it measures the current, realizable worth of the shares, while book value is only history. So even when ROE measures management's financial performance, it may not be synonymous with a high return on investment to shareholders. Thus, it is not enough for investors to find companies capable of generating high ROEs; these companies must be unknown to others, because once they are known, the possibility of high returns to investors will melt away in higher stock prices.

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Part finance, part accounting. An innovative look at the use of accounting information to address selected financial questions, especially business valuation. Now available in paperback.

All of these sites provide vast amounts of information on publicly traded companies, including overviews, stock quotes, financial statements, ratios, charts, and much more.

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PROBLEMS

Answers to odd-numbered problems appear at the end of the book. Answers to even-numbered problems and additional exercises are available in the Instructor Resources within McGraw-Hill's *Connect* (see the Preface for more information).

1. The Board of Directors of Vintage Apparel, Inc., has been pressuring its CEO to boost ROE. During an interview on CNBC, she announced a plan to improve the firm's financial performance. She will raise prices on all of the company's products by 10 percent. She justifies the plan by observing that ROE can be decomposed into the product of profit margin, asset turnover, and financial leverage. By raising prices, she will increase the profit margin and thus ROE. Does this plan make sense to you? Why or why not?
2. a. Which company would you expect to have a higher price-to-earnings ratio: tech giant Alphabet or railroad company Union Pacific? Why?
   b. Which company would you expect to have the higher debt-to-equity ratio: a financial institution or a high-technology company? Why?
   c. Which company would you expect to have a higher profit margin, an appliance manufacturer or a grocer? Why?
   d. Which company would you expect to have a higher current ratio, a jewelry store or an online bookstore? Why?
3. Answer true or false to each of the following. Briefly explain your reasoning for each answer.

a. A company's assets-to-equity ratio always equals one plus its liabilities-to-equity ratio.
b. A company's return on equity will always equal or exceed its return on assets.
c. A company's collection period should always be less than its payables period.
d. A company's current ratio must always equal or exceed its acid-test ratio.
e. All else equal, a firm would prefer to have a higher asset turnover ratio.
f. Two firms can have the same earnings yield but different price-to-earnings ratios.
g. Ignoring taxes and transactions costs, unrealized paper gains are less valuable than realized cash earnings.

4. Your firm is considering the acquisition of a very promising technology company. One executive argues against the move, pointing out that because the technology company is presently losing money, the acquisition will cause your firm's return on equity to fall.
   a. Is the executive correct in predicting that ROE will fall?
   b. How important should changes in ROE be in this decision?
5. Selected financial data for Amberjack Corporation follows.

	($ thousands)	
	Year 1	**Year 2**
Sales	271,161	457,977
Cost of goods sold	249,181	341,204
Net income	(155,034)	(403,509)
Cash flow from operations	(58,405)	(20,437)
Cash	341,180	268,872
Marketable securities	341,762	36,900
Accounts receivable	21,011	35,298
Inventories	6,473	72,106
Total current assets	710,427	413,176
Accounts payable	28,908	22,758
Accrued liabilities	44,310	124,851
Total current liabilities	73,218	147,610

   a. Calculate the current and quick ratio at the end of each year. How has the company's short-term liquidity changed over this period?
   b. Assuming a 365-day year for all calculations, compute the following:
      i. The collection period each year based on sales.
      ii. The inventory turnover and the payables period each year based on cost of goods sold.

iii. The days' sales in cash each year.
iv. The gross margin and profit margin each year.

c. What do these calculations suggest about the company's performance?

6. Top management measures your division's performance by calculating the division's return on investment (ROI), defined as division operating income divided by division assets. Your division has done quite well recently; its ROI is 30 percent. You believe the division should invest in a new production process, but a colleague disagrees, pointing out that because the new investment's first-year ROI is only 25 percent, it will hurt performance. How would you respond?

7. Answer the following questions based on the information in the table. Assume a tax rate of 30 percent. For simplicity, assume that the companies have no other liabilities other than the debt shown. (All dollars are in millions.)

	Atlantic Corp.	**Pacific Corp.**
Earnings before interest and taxes	$450	$ 470
Debt (at 7% interest)	$290	$1,490
Equity	$910	$ 370

a. Calculate each company's ROE, ROA, and ROIC.
b. Why is Pacific's ROE so much higher than Atlantic's? Does this mean Pacific is a better company? Why or why not?
c. Why is Atlantic's ROA higher than Pacific's? What does this tell you about the two companies?
d. How do the two companies' ROICs compare? What does this suggest about the two companies?

8. Table 3.1 in Chapter 3 presents financial statements over the period 2018 through 2021 for R&E Supplies, Inc.
   a. Use these statements to calculate as many of the ratios in Table 2.2 as you can.
   b. What insights do these ratios provide about R&E's financial performance? What problems, if any, does the company appear to have?

9. You are trying to prepare financial statements for Bartlett Pickle Company, but seem to be missing its balance sheet. You have Bartlett's income statement, which shows sales last year were $420 million with a gross profit margin of 40 percent. You also know that credit sales equaled three-quarters of Bartlett's total revenues last year. In addition, Bartlett had a collection period of 55 days, a payables period of 40 days, and an inventory turnover of 8 times based on cost of goods sold. Calculate Bartlett's year-ending balance for accounts receivable, inventory, and accounts payable.

10. In 2020, Natural Selection, a nationwide computer dating service, had $500 million of assets and $200 million of liabilities. Earnings before interest and taxes were $120 million, interest expense was $28 million, the tax rate was 40 percent, principal repayment requirements were $24 million, and annual dividends were 30 cents per share on 20 million shares outstanding.
    a. Calculate the following for Natural Selection:
        i. Liabilities-to-equity ratio
        ii. Times-interest-earned ratio
        iii. Times burden covered
    b. What percentage decline in earnings before interest and taxes could Natural Selection have sustained before failing to cover:
        i. Interest payment requirements
        ii. Principal and interest requirements
        iii. Principal, interest, and common dividend payments
11. Given the following information, complete the balance sheet shown next.

Collection period	71 days
Days' sales in cash	34 days
Current ratio	2.6
Inventory turnover	5 times
Liabilities to assets	75%
Payables period	36 days

(All sales are on credit. All calculations assume a 365-day year. The payables period is based on cost of goods sold.)

Assets	
Current assets:	
Cash	$1,100,000
Accounts receivable	
Inventory	1,900,000
Total current assets	
Net fixed assets	
Total assets	8,000,000
Liabilities and shareholders' equity	
Current liabilities:	
Accounts payable	
Short-term debt	
Total current liabilities	
Long-term debt	
Shareholders' equity	
Total liabilities and equity	

Berk

Fundamentals of Corporate Finance, 2nd Edition {Jonathan Ber

- Balance sheet and balance sheet analysis (market-to-book, D/E, EV): Ch. 2, §2.2–2.3 (p. 26–33)
- Income statement and its analysis (margins, coverage, DuPont, P/E): Ch. 2, §2.4–2.5 (p. 34–42)
- Statement of cash flows: Ch. 2, §2.6 (p. 43–45)

International Financial Reporting Standards

Generally Accepted Accounting Principles (GAAP) differ among countries. As a result, companies face tremendous accounting complexities when they operate internationally. Investors also face difficulty interpreting financial statements of foreign companies, which discourages them from investing abroad. As companies and capital markets become more global, however, interest in harmonization of accounting standards across countries has increased.

The most important harmonization project began in 1973 when representatives of ten countries (including the United States) established the International Accounting Standards Committee. This effort led to the creation of the International Accounting Standards Board (IASB) in 2001, with headquarters in London. Now the IASB has issued a set of International Financial Reporting Standards (IFRS).

The IFRS are taking root throughout the world. The European Union (EU) approved an accounting regulation in 2002 requiring all publicly traded EU companies to follow IFRS in their consolidated financial statements starting in 2005. Many other countries have adopted IFRS for all listed companies, including Australia and several countries in Latin America and Africa. In fact, all major stock exchanges around the world accept IFRS except the United States and Japan, which maintain their local GAAP.

Convergence to IFRS in the United States is likely within the near future. In 2008, the SEC eliminated the requirement for foreign firms listing in U.S. markets to reconcile IFRS to U.S. GAAP. In 2010, the SEC further affirmed its support of a single standard, with IFRS as the preferred method. The SEC will study and revisit the issue in 2011, with the possibility of requiring U.S. companies to start reporting according to IFRS as soon as 2015 or 2016.

Concept Check

1. What is the role of an auditor?
2. What are the four financial statements that all public companies must produce?

2.2 The Balance Sheet

balance sheet A list of a firm's assets and liabilities that provides a snapshot of the firm's financial position at a given point in time.

assets The cash, inventory, property, plant and equipment, and other investments a company has made.

liabilities A firm's obligations to its creditors.

shareholders' equity, stockholders' equity An accounting measure of a firm's net worth that represents the difference between the firm's assets and its liabilities.

common stock and paid-in surplus The amount that stockholders have directly invested in the firm through purchasing stock from the company.

The **balance sheet,** or **statement of financial position**,[2] lists the firm's *assets* and *liabilities,* providing a snapshot of the firm's financial position at a given point in time. Table 2.1 shows the balance sheet for a fictitious company, Global Corporation. Notice that the balance sheet is divided into two parts ("sides") with the assets on the left side and the liabilities on the right side.

1. The **assets** list the firm's cash, inventory, property, plant and equipment, and any other investments the company has made.
2. The **liabilities** show the firm's obligations to its creditors.
3. Also shown with liabilities on the right side of the balance sheet is the *stockholders' equity*. **Stockholders' equity** (also **shareholders' equity**), the difference between the firm's assets and liabilities, is an accounting measure of the firm's net worth. For Global, the stockholders' equity has two parts: (1) **common stock and paid-in surplus**, the amount that stockholders have directly invested in the firm through purchasing stock from the company; and (2) **retained earnings**, which are profits made by the firm, but retained within the firm and reinvested in assets or held as cash. We will take a more detailed look at retained earnings in our discussion of the statement of cash flows later in this chapter.

The assets on the left side show how the firm uses its capital (its investments), and the information on the right side summarizes the sources of capital, or how the firm raises the money it needs. Because of the way stockholders' equity is calculated, the left and right sides must balance:

The Balance Sheet Identity

$$\text{Assets} = \text{Liabilities} + \text{Stockholders' Equity} \tag{2.1}$$

[2]In IFRS and recent U.S. GAAP pronouncements, the balance sheet is referred to as the *statement of financial position*.

TABLE 2.1

Global Corporation Balance Sheet for 2010 and 2009 ($ millions)

Assets	2010	2009	Liabilities and Stockholders' Equity	2010	2009
Current Assets			Current Liabilities		
Cash	23.2	20.5	Accounts payable	29.2	26.5
Accounts receivable	18.5	13.2	Notes payable/short-term debt	5.5	3.2
Inventories	15.3	14.3			
Total current assets	57.0	48.0	Total current liabilities	34.7	29.7
Long-Term Assets			Long-Term Liabilities		
Net property, plant, and equipment	113.1	80.9	Long-term debt	113.2	78.0
Total long-term assets	113.1	80.9	Total long-term liabilities	113.2	78.0
			Total Liabilities	**147.9**	**107.7**
			Stockholders' Equity		
			Common stock and paid-in surplus	8.0	8.0
			Retained earnings	14.2	13.2
			Total Stockholders' Equity	**22.2**	**21.2**
Total Assets	**170.1**	**128.9**	**Total Liabilities and Stockholders' Equity**	**170.1**	**128.9**

retained earnings Profits made by the firm, but retained within the firm and reinvested in assets or held as cash.

current assets Cash or assets that could be converted into cash within one year.

marketable securities Short-term, low-risk investments that can be easily sold and converted to cash.

accounts receivable Amounts owed to a firm by customers who have purchased goods or services on credit.

inventories A firm's raw materials as well as its work-in-progress and finished goods.

long-term assets Assets that produce tangible benefits for more than one year.

In Table 2.1, total assets for 2010 ($170.1 million) are equal to total liabilities ($147.9 million) plus stockholders' equity ($22.2 million).

We now examine the firm's assets, liabilities, and stockholders' equity in more detail. Following this, we evaluate the firm's financial standing by analyzing the information contained in the balance sheet.

Assets

In Table 2.1, Global's assets are divided into *current* and *long-term assets*. We discuss each in turn.

Current Assets. **Current assets** are either cash or assets that could be converted into cash within one year. This category includes:

1. Cash and other **marketable securities**, which are short-term, low-risk investments that can be easily sold and converted to cash (such as money market investments, like government debt, that mature within a year);
2. **Accounts receivable**, which are amounts owed to the firm by customers who have purchased goods or services on credit;
3. **Inventories**, which are composed of raw materials as well as work-in-progress and finished goods; and
4. Other current assets, which is a catch-all category that includes items such as prepaid expenses (such as rent or insurance).

Long-Term Assets. Assets such as real estate or machinery that produce tangible benefits for more than one year are called **long-term assets**. If Global spends $2 million on new equipment, this $2 million will be included with net property, plant, and equipment under long-term assets on the balance sheet. Because equipment tends to wear out or

depreciation A yearly deduction a firm makes from the value of its fixed assets (other than land) over time, according to a depreciation schedule that depends on an asset's life span.

book value The acquisition cost of an asset less its accumulated depreciation.

become obsolete over time, Global will reduce the value recorded for this equipment through a yearly deduction called **depreciation** according to a depreciation schedule that depends on an asset's life span. Depreciation is not an actual cash expense that the firm pays; it is a way of recognizing that buildings and equipment wear out and thus become less valuable the older they get. The **book value** of an asset is equal to its acquisition cost less accumulated depreciation. The figures for net property, plant, and equipment show the total book value of these assets.

Other long-term assets can include such items as property not used in business operations, start-up costs in connection with a new business, trademarks and patents, and property held for sale. The sum of all the firms' assets is the total assets at the bottom of the left side of the balance sheet in Table 2.1.

Liabilities

We now examine the liabilities, shown on the right side of the balance sheet, which are divided into *current* and *long-term liabilities*.

current liabilities Liabilities that will be satisfied within one year.

accounts payable The amounts owed to creditors for products or services purchased with credit.

notes payable, short-term debt Loans that must be repaid in the next year.

Current Liabilities. Liabilities that will be satisfied within one year are known as **current liabilities**. They include:

1. **Accounts payable**, the amounts owed to suppliers for products or services purchased with credit.
2. **Notes payable** and **short-term debt**, loans that must be repaid in the next year. Any repayment of long-term debt that will occur within the next year would also be listed here as current maturities of long-term debt.
3. Accrual items, such as salary or taxes, that are owed but have not yet been paid, and deferred or unearned revenue, which is revenue that has been received for products that have not yet been delivered.

net working capital The difference between a firm's current assets and current liabilities that represents the capital available in the short term to run the business.

The difference between current assets and current liabilities is the firm's **net working capital**, the capital available in the short term to run the business. While notes payable and short-term debt are included in current liabilities, they are different from accounts payable and accrual items. Notes payable and short-term debt are related to financing decisions of the firm, while accounts payable and accruals arise from operating decisions of the firm. This distinction is important later when we see that financial managers generally try to keep these two decisions (operating and financing) separate.

$$\text{Net Working Capital} = \text{Current Assets} - \text{Current Liabilities} \tag{2.2}$$

For example, in 2010 Global's net working capital totaled $22.3 million ($57.0 million in current assets −$34.7 million in current liabilities). Firms with low (or negative) net working capital may face a shortage of funds. In such cases, the liabilities due in the short term exceed the company's cash and expected payments on receivables.

long-term debt Any loan or debt obligation with a maturity of more than a year.

book value of equity The difference between the book value of a firm's assets and its liabilities; also called *shareholders' equity* and *stockholders' equity*, it represents the net worth of a firm from an accounting perspective.

Long-Term Liabilities. Long-term liabilities are liabilities that extend beyond one year. When a firm needs to raise funds to purchase an asset or make an investment, it may borrow those funds through a long-term loan. That loan would appear on the balance sheet as **long-term debt**, which is any loan or debt obligation with a maturity of more than a year.

Stockholders' Equity

The sum of the current liabilities and long-term liabilities is total liabilities. The difference between the firm's assets and liabilities is the *stockholders' equity;* it is also called the **book value of equity** or *shareholders' equity*. As we stated earlier, it represents the net worth of the firm from an accounting perspective. The two main components are com-

mon stock and paid-in surplus and retained earnings. These two components form the book value of stockholders' ownership claims, stemming from their direct investment and reinvestment of profits.

Ideally, the balance sheet would provide us with an accurate assessment of the true value of the firm's equity. Unfortunately, this is unlikely to be the case. First, many of the assets listed on the balance sheet are valued based on their historical cost rather than their true value today. For example, an office building is listed on the balance sheet according to its historical cost less its accumulated depreciation. But the actual value of the office building today may be very different than this amount; in fact, if real estate prices went up it will be worth *more* than the amount the firm paid for it years ago. The same is true for other property, plant, and equipment: The true value today of an asset may be very different from, and even exceed, its book value. A second, and probably more important, problem is that *many of the firm's valuable assets are not captured on the balance sheet*. Consider, for example, the expertise of the firm's employees, the firm's reputation in the marketplace, the relationships with customers and suppliers, and the quality of the management team. All these assets add to the value of the firm but do not appear on the balance sheet.

For these reasons, the book value of equity is an inaccurate assessment of the actual value of the firm's equity. Thus, it is not surprising that it will often differ substantially from the amount investors are willing to pay for the equity. The total market value of a firm's equity equals the market price per share times the number of shares, referred to as the company's **market capitalization**. The market value of a stock does not depend on the historical cost of the firm's assets; instead, it depends on what investors expect those assets to produce in the future. To see the difference, think about what happens when a company like Boeing unveils a new plane. Investors' expectations about future cash flows from selling those planes increase the stock price immediately, elevating the market value of Boeing. However, the revenue from selling the planes will only be reflected in Boeing's financial statements when it actually sells them.

market capitalization
The total market value of equity; equals the market price per share times the number of shares.

EXAMPLE 2.1

Market Versus Book Value

Problem

If Global has 3.6 million shares outstanding, and these shares are trading for a price of $10.00 per share, what is Global's market capitalization? How does the market capitalization compare to Global's book value of equity?

Solution

Plan

Market capitalization is equal to price per share times shares outstanding. We can find Global's book value of equity at the bottom of the right side of its balance sheet.

Execute

Global's market capitalization is (10.00/share) × (3.6 million shares) = $36 million. This market capitalization is significantly higher than Global's book value of equity of $22.2 million.

Evaluate

Global must have sources of value that do not appear on the balance sheet. These include potential opportunities for growth, the quality of the management team, relationships with suppliers and customers, etc.

Finally, we note that the book value of equity can be negative (liabilities exceed assets), and that a negative book value of equity is not necessarily an indication of poor performance. Successful firms are often able to borrow in excess of the book value of their assets because creditors recognize that the market value of the assets is far higher. For example, in June 2005 Amazon.com had total liabilities of $2.6 billion and a book value of equity of −$64 million. At the same time, the market value of its equity was over $15 billion. Clearly,

investors recognized that Amazon's assets were worth far more than the book value reported on the balance sheet. By 2010, several years of strong growth had brought its book equity to over $5 billion and its market value of equity to more than $50 billion!

Concept Check

3. What is depreciation designed to capture?
4. The book value of a company's assets usually does not equal the market value of those assets. What are some reasons for this difference?

2.3 Balance Sheet Analysis

What can we learn from analyzing a firm's balance sheet? Although the book value of a firm's equity is not a good estimate of its true value as an ongoing firm, it is sometimes used as an estimate of the **liquidation value** of the firm, the value that would be left after its assets were sold and liabilities paid. We can also learn a great deal of useful information from a firm's balance sheet that goes beyond the book value of the firm's equity. We now discuss analyzing the balance sheet to assess the firm's value, its leverage, and its short-term cash needs.

liquidation value The value of a firm after its assets are sold and liabilities paid.

Market-to-Book Ratio

In Example 2.1, we compared the market and book values of Global's equity. A common way to make this comparison is to compute the **market-to-book ratio** (also called the **price-to-book [P/B] ratio**), which is the ratio of a firm's market capitalization to the book value of stockholders' equity.

market-to-book ratio (price-to-book [P/B] ratio) The ratio of a firm's market (equity) capitalization to the book value of its stockholders' equity.

$$\text{Market-to-Book Ratio} = \frac{\text{Market Value of Equity}}{\text{Book Value of Equity}} \tag{2.3}$$

It is one of many financial ratios used to evaluate a firm. The market-to-book ratio for most successful firms substantially exceeds 1, indicating that the value of the firm's assets when put to use exceeds their historical cost (or liquidation value). The ratio will vary across firms due to differences in fundamental firm characteristics as well as the value added by management. Thus, this ratio is one way a company's stock price provides feedback to its managers on the market's assessment of their decisions.

In early 2010, AmeriServ Financial, Inc. (ASRV) had a market-to-book ratio of 0.52, a reflection of investors' assessment that many of AmeriServ's assets were unlikely to be profitable and were worth less than their book value. Citigroup's market-to-book ratio of 0.82 tells a similar story. Figure 2.1 shows that at the same time, the average market-to-book ratio for the financial services industry was about 1, and for large U.S. firms it was close to 3. In contrast, consider that Amazon.com, Inc. (AMZN) had a market-to-book ratio of over 11, and the average for technology firms was about 6. Analysts often classify firms with low market-to-book ratios as **value stocks**, and those with high market-to-book ratios as **growth stocks**.

value stocks Firms with low market-to-book ratios.

growth stocks Firms with high market-to-book ratios.

Debt-Equity Ratio

Another important piece of information that we can learn from a firm's balance sheet is the firm's **leverage**, or the extent to which it relies on debt as a source of financing. The **debt-equity ratio** is a common ratio used to assess a firm's leverage that we calculate by dividing the total amount of short- and long-term debt (including current maturities) by the total stockholders' equity:

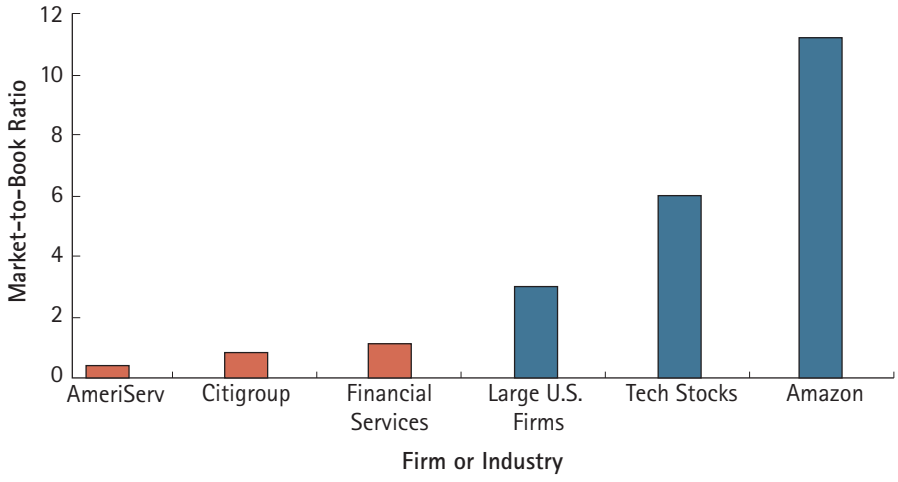
leverage A measure of the extent to which a firm relies on debt as a source of financing.

debt-equity ratio The ratio of a firm's total amount of short- and long-term debt (including current maturities) to the value of its equity, which may be calculated based on market or book values.

$$\text{Debt-Equity Ratio} = \frac{\text{Total Debt}}{\text{Total Equity}} \tag{2.4}$$

FIGURE 2.1
Market-to-Book Ratios in 2010

This figure presents market-to-book ratios of different firms and groups of firms in 2010. Firms that might be classified as value stocks (low market-to-book ratios) are in red and those that might be classified as growth stocks (high market-to-book ratios) are in blue.



We can calculate this ratio using either book or market values for equity and debt. From Table 2.1, Global's debt in 2010 includes notes payable ($5.5 million) and long-term debt ($113.2 million), for a total of $118.7 million. Therefore, using the book value of equity, its *book* debt-equity ratio is 118.7/22.2 = 5.3. Note the large increase from 2009, when the book debt-equity ratio was only (3.2 + 78)/21.2 = 3.8.

Because of the difficulty interpreting the book value of equity, the book debt-equity ratio is not especially useful. It is more informative to compare the firm's debt to the market value of its equity. Global's debt-equity ratio in 2010, using the market value of equity (from Example 2.1), is 118.7/36 = 3.3, which means Global's debt is a bit more than triple the market value of its equity.[3] As we will see later in the text, a firm's *market* debt-equity ratio has important consequences for the risk and return of its stock.

Enterprise Value

enterprise value The total market value of a firm's equity and debt, less the value of its cash and marketable securities. It measures the value of the firm's underlying business.

A firm's market capitalization measures the market value of the firm's equity, or the value that remains after the firm has paid its debts. So it includes any cash the firm holds. But what if you are interested in just the value of the business itself? The **enterprise value** of a firm assesses the value of the underlying business assets, unencumbered by debt and separate from any cash and marketable securities. We compute it as follows:

$$\text{Enterprise Value} = \text{Market Value of Equity} + \text{Debt} - \text{Cash} \tag{2.5}$$

For example, given its market capitalization from Example 2.1, Global's enterprise value in 2010 is 36 + 118.7 − 23.2 = $131.5 million. We can interpret the enterprise value as the cost to take over the business. That is, it would cost 36 + 118.7 = $154.7 million to buy all of Global's equity and pay off its debts. Because we would acquire Global's $23.2 million in cash (which can be used to pay off Global's debt), the net cost is only 154.7 − 23.2 = $131.5 million.

[3]In this calculation, we have compared the market value of equity to the book value of debt. Strictly speaking, it would be best to use the market value of debt. But because the market value of debt is generally not very different from its book value, this distinction is often ignored in practice.

EXAMPLE 2.2
Computing Enterprise Value

Problem

In April 2010, H. J. Heinz Co. (HNZ) had a share price of $46.15, 316.2 million shares outstanding, a market-to-book ratio of 7.99, a book debt-equity ratio of 2.64, and cash of $562.3 million. What was Heinz's market capitalization (its market value of equity)? What was its enterprise value?

Solution

Plan

Share Price	$46.15
Shares Outstanding	316.2 million
Market-to-Book	7.99
Cash	$562.3 million
Debt-to-Equity (Book)	2.64

We will solve the problem using Eq. 2.5: Enterprise Value = Market Value of Equity + Debt − Cash. We can compute the market capitalization by multiplying the share price by the shares outstanding. We are given the amount of cash. We are not given the debt directly, but we are given the book debt-to-equity ratio. If we knew the book value of equity, we could use the ratio to infer the value of the debt. Since we can compute the market value of equity (market capitalization) and we have the market-to-book ratio, we can compute the book value of equity, so that is the last piece of information we will need.

Execute

Heinz had a market capitalization of $46.15/share × 316.2 million shares = $14.59 billion. Since Heinz's market-to-book = 7.99 = $14.59 billion/book equity, then book equity = $14.59 billion/7.99 = $1.83 billion. Given that the book equity is $1.83 billion and a book debt-to-equity ratio of 2.64, the total value of Heinz's debt is $1.83 billion × 2.64 = $4.83 billion.

Evaluate

Thus, Heinz's enterprise value was 14.59 + 4.83 − 0.5623 = $18.858 billion.

Other Balance Sheet Information

current ratio The ratio of current assets to current liabilities.

quick ratio The ratio of current assets other than inventory to current liabilities.

Creditors often compare a firm's current assets and current liabilities to assess whether the firm has sufficient working capital to meet its short-term needs. This comparison is sometimes summarized in the firm's **current ratio**, the ratio of current assets to current liabilities, or its **quick ratio ("acid-test" ratio)**, the ratio of current assets other than inventory to current liabilities. A higher current or quick ratio implies less risk of the firm experiencing a cash shortfall in the near future.

$$\text{Current Ratio} = \frac{\text{Current Assets}}{\text{Current Liabilities}} \tag{2.6}$$

$$\text{Quick Ratio} = \frac{\text{Current Assets} - \text{Inventory}}{\text{Current Liabilities}} \tag{2.7}$$

Analysts also use the information on the balance sheet to watch for trends that could provide information regarding the firm's future performance. For example, an unusual increase in inventory could be an indicator that the firm is having difficulty selling its products.

Table 2.2 summarizes balance sheet ratios and provides typical values of those ratios during an economic expansion as well as what happened to those ratios during the recent financial crisis and subsequent recession. Many of the recession effects noted here and in future tables are generally short-term in nature because once it is clear that the economy is in a recession, firms take actions to mitigate these effects. The table gives average values for the manufacturing, retail, and service sectors along with the 500 firms in the S&P

TABLE 2.2
Balance Sheet Ratios

Ratio	Formula	Manu-facturing	Retail	Service	S&P 500	Financial Crisis and Recession
Market-to-Book Ratio	$\frac{\text{Market Value of Equity}}{\text{Book Value of Equity}}$	2.53	2.40	2.73	2.31	↓
Book Debt-to-Equity Ratio	$\frac{\text{Total Debt}}{\text{Book Value of Total Equity}}$	18.9%	31.7%	8.7%	32.4%	↑
Market Debt-to-Equity Ratio	$\frac{\text{Total Debt}}{\text{Market Value of Total Equity}}$	7.1%	14.8%	2.5%	13.2%	↑↑
Current Ratio	$\frac{\text{Current Assets}}{\text{Current Liabilities}}$	2.31	1.51	1.52	1.47	—
Quick Ratio	$\frac{\text{Current Assets} - \text{Inventory}}{\text{Current Liabilities}}$	1.59	0.73	1.43	1.14	—

Notes: A "—" in the recession column indicates that any changes in the ratio during the recession were too small to be economically significant or were not broadly felt across sectors. Two arrows indicate a larger move than a single arrow.

Source: *Standard and Poors' Compustat.*

500 index. The market-to-book ratio is an indicator of potential growth and of managers' ability to generate value from the firm's assets above their historical cost. The other ratios measure the financial health of the firm by assessing its leverage (debt-to-equity and equity multiplier) or liquidity (current and quick ratios).

Concept Check

5. What does a high debt-to-equity ratio tell you?
6. What is a firm's enterprise value?

2.4 The Income Statement

income statement A list of a firm's revenues and expenses over a period of time.

net income or earnings The last or "bottom" line of a firm's income statement that is a measure of the firm's income over a given period of time.

When you want someone to get to the point, you might ask them for the "bottom line." This expression comes from the *income statement*. The **income statement** lists the firm's revenues and expenses over a period of time. The last or "bottom" line of the income statement shows the firm's **net income**, which is a measure of its profitability during the period. The income statement is sometimes called a *profit and loss* or *P&L statement*, and the net income is also referred to as the firm's **earnings**. In this section, we examine the components of the income statement in detail and introduce ratios we can use to analyze this data.

Earnings Calculations

Whereas the balance sheet shows the firm's assets and liabilities at a given point in time, the income statement shows the flow of revenues and expenses generated by those assets and liabilities between two dates. Table 2.3 shows Global's income statement for 2010 and 2009. We examine each category on the statement.

gross profit The third line of an income statement that represents the difference between a firm's sales revenues and its costs.

Gross Profit. The first two lines of the income statement list the revenues from sales of products and the costs incurred to make and sell the products. Note that in accounting, the terms *revenues* and *net sales* are equivalent. Net sales is simply gross sales minus any returns, discounts, and allowances. We will simply use the term *sales* from here forward. The third line is **gross profit**, the difference between sales revenues and the costs.

TABLE 2.3
Global Corporation's Income Statement Sheet for 2010 and 2009

GLOBAL CORPORATION
Income Statement
Year ended December 31 (in $ millions)

	2010	2009
Net sales	186.7	176.1
Cost of sales	–153.4	–147.3
Gross Profit	33.3	28.8
Selling, general, and administrative expenses	–13.5	–13
Research and development	–8.2	–7.6
Depreciation and amortization	–1.2	–1.1
Operating Income	10.4	7.1
Other income	—	—
Earnings Before Interest and Taxes (EBIT)	10.4	7.1
Interest income (expense)	–7.7	–4.6
Pretax Income	2.7	2.5
Taxes	–0.7	–0.6
Net Income	2.0	1.9
Earnings per share:	$0.56	$0.53
Diluted earnings per share:	$0.53	$0.50

Operating Expenses. The next group of items is operating expenses. These are expenses from the ordinary course of running the business that are not directly related to producing the goods or services being sold. They include administrative expenses and overhead, salaries, marketing costs, and research and development expenses. The third type of operating expense, depreciation and amortization (a charge that captures the change in value of acquired assets), represents an estimate of the costs that arise from wear and tear or obsolescence of the firm's assets.[4] It is not an actual cash expense. The firm's gross profit net of operating expenses is called **operating income**.

operating income A firm's gross profit less its operating expenses.

Earnings Before Interest and Taxes. We next include other sources of income or expenses that arise from activities that are not the central part of a company's business. Cash flows from the firm's financial investments are one example of other income that would be listed here. After we have adjusted for other sources of income or expenses, we have the firm's earnings before interest and taxes, or **EBIT**.

EBIT A firm's earnings before interest and taxes are deducted.

Pretax and Net Income. From EBIT, we deduct the interest paid on outstanding debt to compute Global's pretax income, and then we deduct corporate taxes to determine the firm's net income.

Net income represents the total earnings of the firm's equity holders. It is often reported on a per-share basis as the firm's **earnings per share (EPS)**, which we compute by dividing net income by the total number of shares outstanding:

earnings per share (EPS) A firm's net income divided by the total number of shares outstanding.

$$\text{EPS} = \frac{\text{Net Income}}{\text{Shares Outstanding}} = \frac{\$2.0 \text{ million}}{3.6 \text{ million shares}} = \$0.56 \text{ per share} \qquad (2.8)$$

[4]Only certain types of amortization are deductible as a pretax expense (e.g., amortization of the cost of an acquired patent). Amortization of goodwill is not a pretax expense and is generally included as an extraordinary item after taxes are deducted.

stock options The right to buy a certain number of shares of stock by a specific date at a specific price.

convertible bonds Corporate bonds with a provision that gives the bondholder an option to convert each bond owned into a fixed number of shares of common stock.

dilution An increase in the total number of shares that will divide a fixed amount of earnings.

Although Global has only 3.6 million shares outstanding as of the end of 2010 (from Example 2.1), the number of shares outstanding may grow if Global has made commitments that would cause it to issue more shares. Consider these two examples:

1. Suppose Global compensates its employees or executives with **stock options** that give the holder the right to buy a certain number of shares by a specific date at a specific price. If employees "exercise" these options, the company issues new stock and the number of shares outstanding will grow.
2. The number of shares may also grow if the firm issues **convertible bonds**, a form of debt that can be converted into shares of common stock.

In the cases of stock options and convertible bonds, because there will be more total shares to divide the same earnings, this growth in the number of shares is referred to as **dilution**. Firms disclose the potential for dilution from options they have awarded by reporting **diluted EPS**, which shows the earnings per share the company would have if the stock options were exercised. For example, if Global has awarded options for 200,000 shares of stock to its key executives, its diluted EPS is \$2.0 million/3.8 million shares = \$0.53.

Concept Check

7. What do a firm's earnings measure?
8. What is dilution?

2.5 Income Statement Analysis

diluted EPS The earnings per share a company would have based on the total number of shares including the effects of all stock options and convertible bonds.

The income statement provides very useful information regarding the profitability of a firm's business and how it relates to the value of the firm's shares. We now discuss several ratios that are often used to evaluate a firm's performance and value.

Profitability Ratios

We introduce three profitability ratios: *gross margin*, *operating margin*, and *net profit margin*.

gross margin The ratio of gross profit to revenues (sales), it reflects the ability of the company to sell a product for more than the sum of the direct costs of making it.

Gross Margin. The **gross margin** of a firm is the ratio of gross profit to revenues (sales):

$$\text{Gross Margin} = \frac{\text{Gross Profit}}{\text{Sales}} \tag{2.9}$$

The gross margin simply reflects the ability of the company to sell a product for more than the sum of the direct costs of making it. All of the firm's other expenses of doing business (those not directly related to producing the goods sold) must be covered by this margin. In 2010 Global's gross profit was \$33.3 million and its sales were \$186.7 million, for a gross margin of 33.3/186.7 = 17.84%.

operating margin The ratio of operating income to revenues, it reveals how much a company has earned from each dollar of sales before deducting interest and taxes.

Operating Margin. Because operating income reflects all of the expenses of doing business, another important profitability ratio is the **operating margin**, the ratio of operating income to revenues:

$$\text{Operating Margin} = \frac{\text{Operating Income}}{\text{Sales}} \tag{2.10}$$

The operating margin reveals how much a company earns before interest and taxes from each dollar of sales. Global's operating margin in 2010 was 10.4/186.7 = 5.57%, an increase from its 2009 operating margin of 7.1/176.1 = 4.03%. By comparing operating margins across firms within an industry, we can assess the relative efficiency of firms'

operations. For example, in 2010 American Airlines (AMR) had an operating margin of −5.04% (i.e., they lost 5 cents for each dollar in revenues). However, competitor Southwest Airlines (LUV) had an operating margin of 2.53%.

Differences in operating margins can also result from differences in strategy. For example, in 2010, Wal-Mart Stores had an operating margin of 5.87% while high-end retailer Nordstrom had an operating margin of 9.67%. In this case, Wal-Mart's lower operating margin is not a result of its inefficiency but is part of its strategy of offering lower prices to sell common products in high volume. Indeed, Wal-Mart's sales were more than 45 times higher than those of Nordstrom.

net profit margin The ratio of net income to revenues, it shows the fraction of each dollar in revenues that is available to equity holders after the firm pays its expenses, plus interest and taxes.

Net Profit Margin. A firm's **net profit margin** is the ratio of net income to revenues:

$$\text{Net Profit Margin} = \frac{\text{Net Income}}{\text{Sales}} \tag{2.11}$$

The net profit margin shows the fraction of each dollar in revenues that is available to equity holders after the firm pays its expenses, plus interest and taxes. Global's net profit margin in 2010 was 2.0/186.7 = 1.07%. Differences in net profit margins can be due to differences in efficiency, but they can also result from differences in leverage (the firm's reliance on debt financing), which determines the amount of interest payments.

Asset Efficiency

A financial manager can use the combined information in the firm's income statement and balance sheet to gauge how efficiently his or her firm is utilizing its assets. A first broad measure of efficiency is asset turnover, the ratio of sales to total assets:

$$\text{Asset Turnover} = \frac{\text{Sales}}{\text{Total Assets}} \tag{2.12}$$

Low values of asset turnover indicate that the firm is not generating much revenue (sales) per dollar of assets. In 2010 Global's $170.1 million in assets generated $186.7 million in sales, for an asset turnover ratio of 1.1 (=$186.7/$170.1). Since total assets includes assets, such as cash, that are not directly involved in generating sales, Global's manager might also look at Global's fixed asset turnover, which is equal to sales divided by fixed assets:

$$\text{Fixed Asset Turnover} = \frac{\text{Sales}}{\text{Fixed Assets}} \tag{2.13}$$

Global's fixed assets in 2010 were $113.1 million worth of property, plant, and equipment, yielding a fixed asset turnover of 1.7 (=$186.7/$113.1). Low asset turnover ratios indicate that the firm is generating relatively few sales given the amount of assets it employs.

Working Capital Ratios

Global's managers might be further interested in how efficiently they are managing their net working capital. We can express the firm's accounts receivable in terms of the number of days' worth of sales that it represents, called the **accounts receivable days, average collection period**, or **days sales outstanding**:[5]

accounts receivable days (average collection period or days sales outstanding) An expression of a firm's accounts receivable in terms of the number of days' worth of sales that the accounts receivable represents.

$$\text{Accounts Receivable Days} = \frac{\text{Accounts Receivable}}{\text{Average Daily Sales}} \tag{2.14}$$

[5]Accounts receivable days can also be calculated based on the average accounts receivable at the end of the current and prior years.

accounts payable days An expression of a firm's accounts payable in terms of the number of days' worth of cost of goods sold that the accounts payable represents.

inventory days An expression of a firm's inventory in terms of the number of days' worth or cost of goods sold that the inventory represents.

inventory turnover ratio The cost of goods sold divided by either the latest cost of inventory or the average inventory over the year, it shows how efficiently companies turn their inventory into sales.

Given average daily sales of \$186.7 million/365 = \$0.51 million in 2010, Global's receivables of \$18.5 million represent 18.5/0.51 = 36 days' worth of sales. In other words, Global takes a little over one month to collect payment from its customers, on average. In 2009, Global's accounts receivable represented only 27.5 days worth of sales. Although the number of receivable days can fluctuate seasonally, a significant unexplained increase could be a cause for concern (perhaps indicating the firm is doing a poor job collecting from its customers or is trying to boost sales by offering generous credit terms). Similar ratios exist for accounts payable and inventory. Those ratios are called **accounts payable days** (accounts payable divided by average daily cost of goods sold) and **inventory days** (inventory divided by average daily cost of goods sold).

We can also compute how efficiently firms use inventory. The **inventory turnover ratio** is equal to the cost of goods sold divided by either the latest cost of inventory or the average inventory over the year. We use the cost of goods sold because that is how inventory costs are reflected on the income statement. It is also common to use sales in the numerator to make the ratio more like the asset turnover ratios.

$$\text{Inventory Turnover} = \frac{\text{Cost of Goods Sold}}{\text{Inventory}} \tag{2.15}$$

A normal level for this ratio, similar to the others in this section, can vary substantially for different industries, although a higher level (more dollars of sales per dollar of inventory) is generally better.

EBITDA

EBITDA A computation of a firm's earnings before interest, taxes, depreciation, and amortization are deducted.

Financial analysts often compute a firm's earnings before interest, taxes, depreciation, and amortization, or **EBITDA**. Because depreciation and amortization are not cash expenses for the firm, EBITDA reflects the cash a firm has earned from its operations. Global's EBITDA in 2010 was 10.4 + 1.2 = \$11.6 million.

Leverage Ratios

interest coverage ratio or times interest earned (TIE) ratio An assessment by lenders of a firm's leverage, it is equal to a measure of earnings divided by interest.

Lenders often assess a firm's leverage by computing an **interest coverage ratio**, also known as a **times interest earned (TIE) ratio**, which is equal to a measure of earnings divided by interest. Financial managers watch these ratios carefully because they assess how easily the firm will be able to cover its interest payments. There is no one accepted measure of earnings for these ratios; it is common to consider operating income, EBIT, or EBITDA as a multiple of the firm's interest expenses. When this ratio is high, it indicates that the firm is earning much more than is necessary to meet its required interest payments.

Investment Returns

return on equity (ROE) The ratio of a firm's net income to the book value of its equity.

Analysts and financial managers often evaluate the firm's return on investment by comparing its income to its investment using ratios such as the firm's **return on equity (ROE)**:[6]

$$\text{Return on Equity} = \frac{\text{Net Income}}{\text{Book Value of Equity}} \tag{2.16}$$

Global's ROE in 2010 was 2.0/22.2 = 9.0%. The ROE provides a measure of the return that the firm has earned on its past investments. A high ROE may indicate the firm

[6]Because net income is measured over the year, the ROE can also be calculated based on the average book value of equity at the end of the current and prior years.

is able to find investment opportunities that are very profitable. Of course, one weakness of this measure is the difficulty in interpreting the book value of equity.

return on assets (ROA) The ratio of net income to the total book value of the firm's assets.

Another common measure is the **return on assets (ROA)**, which is net income divided by the total assets. A firm must earn both a positive ROE and ROA to grow.

The DuPont Identity

Global's financial manager will need to know that its ROE is 9%, but that financial manager would also need to understand the drivers of his or her firm's return on equity. High margins, efficient use of assets, or even simply high leverage could all lead to a higher return on equity. By delving deeper into the sources of return on equity, the financial manager can gain a clear sense of the firm's financial picture. One common tool for doing so is the **DuPont Identity**, named for the company that popularized it, which expresses return on equity as the product of profit margin, asset turnover, and a measure of leverage.

DuPont Identity Expresses return on equity as the product of profit margin, asset turnover, and a measure of leverage.

To understand the DuPont Identity, we start with ROE and decompose it in steps into the drivers identified in the identity. First, we simply multiply ROE by (sales/sales), which is just 1, and rearrange terms:

$$\text{ROE} = \left(\frac{\text{Net Income}}{\text{Total Equity}}\right)\left(\frac{\text{Sales}}{\text{Sales}}\right) = \left(\frac{\text{Net Income}}{\text{Sales}}\right)\left(\frac{\text{Sales}}{\text{Total Equity}}\right) \tag{2.17}$$

This expression says that ROE can be thought of as net income per dollar of sales (profit margin) times the amount of sales per dollar of equity. For example, Global's ROE comes from its profit margin of 1.07% multiplied by its sales per dollar of equity $(186.7/22.2 = 8.41)$: $1.07\% \times 8.41 = 9\%$. We can take the decomposition further by multiplying Eq. 2.17 by (total assets / total assets), which again is just 1, and rearranging the terms:

DuPont Identity

$$\begin{aligned}\text{ROE} &= \left(\frac{\text{Net Income}}{\text{Sales}}\right)\left(\frac{\text{Sales}}{\text{Total Equity}}\right)\left(\frac{\text{Total Assets}}{\text{Total Assets}}\right)\\ &= \left(\frac{\text{Net Income}}{\text{Sales}}\right)\left(\frac{\text{Sales}}{\text{Total Assets}}\right)\left(\frac{\text{Total Assets}}{\text{Total Equity}}\right)\end{aligned} \tag{2.18}$$

This final expression says that ROE is equal to net income per dollar of sales (profit margin) times sales per dollar of assets (asset turnover) times assets per dollar of equity (a measure of leverage called the **equity multiplier**). Equation 2.18 is the DuPont Identity, expressing return on equity as the product of profit margin, asset turnover, and the equity multiplier. Turning to Global, its equity multiplier is $7.7 (=170.1/22.2)$. A financial manager at Global looking for ways to increase ROE could assess the drivers behind its current ROE with the DuPont Identity. With a profit margin of 1.07%, asset turnover of 1.1, and an equity multiplier of 7.7, we have:

equity multiplier A measure of leverage equal to total assets divided by total equity.

$$\text{ROE} = 9\% = (1.07\%)(1.1)(7.7)$$

This decomposition of ROE shows that leverage is already high (confirmed by the fact that the book debt-to-equity ratio shows that Global's debt is more than five times its equity). However, Global is operating with only 1% profit margins and relatively low asset turnover. Thus, Global's managers should focus on utilizing the firm's existing assets more efficiently and lowering costs to increase the profit margin.[7]

[7]Although the DuPont Identity makes it look like you can increase ROE just by increasing leverage, it is not quite that simple. An increase in leverage will increase your interest expense, decreasing your profit margin.

EXAMPLE 2.3
DuPont Analysis

Problem

The following table contains information about Wal-Mart (WMT) and Nordstrom (JWN). Compute their respective ROEs and then determine how much Wal-Mart would need to increase its profit margin in order to match Nordstrom's ROE.

	Profit Margin	Asset Turnover	Equity Multiplier
Wal-Mart	3.6%	2.4	2.6
Nordstrom	7.7%	1.7	2.4

Solution

Plan and Organize

The table contains all the relevant information to use the DuPont Identity to compute the ROE. We can compute the ROE of each company by multiplying together its profit margin, asset turnover, and equity multiplier. In order to determine how much Wal-Mart would need to increase its profit margin to match Nordstrom's ROE, we can set Wal-Mart's ROE equal to Nordstrom's, keep its turnover and equity multiplier fixed, and solve for the profit margin.

Execute

Using the DuPont Identity, we have:

$$\text{ROE}_{\text{WMT}} = 3.6\% \times 2.4 \times 2.6 = 22.5\%$$
$$\text{ROE}_{\text{JWN}} = 7.7\% \times 1.7 \times 2.4 = 31.4\%$$

Now, using Nordstrom's ROE, but Wal-Mart's asset turnover and equity multiplier, we can solve for the profit margin that Wal-Mart needs to achieve Nordstrom's ROE:

$$31.4\% = \text{Margin} \times 2.4 \times 2.6$$
$$\text{Margin} = 31.4\%/6.24 = 5.0\%$$

Evaluate

Wal-Mart would have to increase its profit margin from 3.6% to 5% in order to match Nordstrom's ROE. It would be able to achieve Nordstrom's ROE even with a lower profit margin than Nordstrom (5.0% vs. 7.7%) because of its higher turnover and slightly higher leverage.

Valuation Ratios

Analysts and investors use a number of ratios to gauge the market value of the firm. The most important is the firm's **price-earnings ratio (P/E)**:

price-earnings ratio (P/E) The ratio of the market value of equity to the firm's earnings, or its share price to its earnings per share.

$$\text{P/E Ratio} = \frac{\text{Market Capitalization}}{\text{Net Income}} = \frac{\text{Share Price}}{\text{Earnings per Share}} \tag{2.19}$$

That is, the P/E ratio is the ratio of the value of equity to the firm's earnings, either on a total basis or on a per-share basis. Following Eq. 2.19, Global's P/E ratio in 2010 was $36/2.0 = 10/0.56 = 18$. The P/E ratio is a simple measure that is used to assess whether a stock is over- or under-valued, based on the idea that the value of a stock should be proportional to the level of earnings it can generate for its shareholders. P/E ratios can vary widely across industries and tend to be higher for industries with high growth rates. For example, in 2010 the average large U.S. firm had a P/E ratio of about 18. But airlines, which had very low current earnings due to the recession, but the promise of higher future earnings as the economy recovered, had an average P/E ratio of 31. One way to capture the idea that a higher P/E ratio can be justified by a higher growth rate is to compare it to the company's expected earnings growth rate. For example, if Global's expected growth rate is 18%, then it would have a P/E to Growth, or **PEG ratio**, of 1. Some investors consider PEG ratios of 1 or below as indicating the stock is fairly priced, but would question whether the company is potentially overvalued if the PEG is higher than 1.

PEG ratio The ratio of a firm's P/E to its expected earnings growth rate.

The P/E ratio considers the value of the firm's equity and so depends on its leverage. Recall that the amount of assets controlled by the equity holders can be increased through the use of leverage. To assess the market value of the underlying business, it is common to consider valuation ratios based on the firm's enterprise value. Typical ratios include the ratio of enterprise value to revenue, or enterprise value to operating income or EBITDA. These ratios compare the value of the business to its sales, operating profits, or cash flow. Similar to the P/E ratio, managers use these ratios to make intra-industry comparisons of how firms are priced in the market.

COMMON MISTAKE

Mismatched Ratios

When considering valuation (and other) ratios, be sure that the items you are comparing both represent amounts related to the entire firm or that both represent amounts related solely to equity holders. For example, a firm's share price and market capitalization are values associated with the firm's equity. Thus, it makes sense to compare them to the firm's earnings per share or net income, which are amounts to equity holders after interest has been paid to debt holders. We must be careful, however, if we compare a firm's market capitalization to its revenues, operating income, or EBITDA. These amounts are related to the whole firm, and both debt and equity holders have a claim to them. Therefore, it is better to compare revenues, operating income, or EBITDA to the enterprise value of the firm, which includes both debt and equity.

The P/E ratio is not useful when the firm's earnings are negative. In this case, it is common to look at the firm's enterprise value relative to sales. The risk in doing so, however, is that earnings might be negative because the firm's underlying business model is fundamentally flawed, as was the case for many Internet firms in the late 1990s.

EXAMPLE 2.4

Computing Profitability and Valuation Ratios

Problem

Consider the following data from 2010 for Wal-Mart Stores and Target Corporation ($ billions):

	Wal-Mart Stores (WMT)	Target Corporation (TGT)
Sales	408	65
Operating Income	24	5
Net Income	14	2
Market Capitalization	203	41
Cash	8	2
Debt	41	17

Compare Wal-Mart and Target's operating margin, net profit margin, P/E ratio, and the ratio of enterprise value to operating income and sales.

Solution

Plan

The table contains all of the raw data, but we need to compute the ratios using the inputs in the table.

Operating Margin = Operating Income/Sales

Net Profit Margin = Net Income/Sales

P/E ratio = Price/Earnings = Market Capitalization/Net Income

Enterprise value to operating income = Enterprise Value/Operating Income

Enterprise value to sales = Enterprise Value/Sales

Execute

Wal-Mart had an operating margin of 24/408 = 5.9%, a net profit margin of 14/408 = 3.4%, and a P/E ratio of 203/14 = 14.5. Its enterprise value was 203 + 41 − 8 = $236 billion. Its ratio to operating income is 236/24 = 9.8, and its ratio to sales is 236/408 = 0.58.

Target had an operating margin of 5/65 = 7.7%, a net profit margin of 2/65 = 3.1%, and a P/E ratio of 41/2 = 20.5. Its enterprise value was 41 + 17 − 2 = $56 billion. Its ratio to operating income is 56/5 = 11.2, and its ratio to sales is 56/65 = 0.86.

Evaluate

Despite a slightly lower overall profit margin, Target has higher P/E and enterprise value ratios, suggesting that the market expects greater growth from Target than from Wal-Mart.

Table 2.4 summarizes income statement ratios and provides typical values of those ratios during an economic expansion as well as what happened to them during the Financial Crisis and ensuing recession.

TABLE 2.4
Income Statement Ratios

Ratio	Formula	Manufacturing	Retail	Service	S&P 500	Financial Crisis and Recession
Profitability Ratios						
Gross margin	$\frac{\text{Gross Profit}}{\text{Sales}}$	34.3%	30.8%	50.4%	38.4%	—
Operating margin	$\frac{\text{Operating Income}}{\text{Sales}}$	8.4%	7.4%	8.7%	19.7%	↓
Net Profit margin	$\frac{\text{Net Income}}{\text{Sales}}$	2.0%	2.3%	2.1%	8.7%	↓↓
Leverage Ratio						
Interest coverage ratio (TIE)	$\frac{\text{Operating Income}}{\text{Interest Expense}}$	4.78	7.16	3.58	12.13	↓
Investment Return Ratios						
Return on equity	$\frac{\text{Net Income}}{\text{Book Value of Equity}}$	7.9%	10.6%	7.9%	15.8%	↓
Return on assets	$\frac{\text{Net Income}}{\text{Total Assets}}$	1.6%	4.3%	1.1%	5.4%	↓↓
Valuation Ratio						
Price-to-earnings ratio	$\frac{\text{Share Price}}{\text{Earnings per Share}}$	10.0	15.2	9.3	18.0	↓*
Efficiency and Working Capital Ratios						
Accounts receivable days	$\frac{\text{Accounts Receivable}}{\text{Average Daily Sales}}$	56.8	6.7	62.5	57.5	—
Fixed asset turnover	$\frac{\text{Sales}}{\text{Fixed Assets}}$	5.6	6.3	11.8	5.2	—
Total asset turnover	$\frac{\text{Sales}}{\text{Total Assets}}$	0.9	1.8	0.8	0.7	—
Inventory turnover	$\frac{\text{Cost of Goods Sold}}{\text{Inventory}}$	4.2	6.5	21.5	6.2	—

Notes: A "—" in the recession column indicates that any changes in the ratio during the recession were too small to be economically significant or were not broadly felt across sectors. Two arrows indicate a larger move than a single arrow.

* Because many firms had losses (negative earnings) during the recession, the P/E ratio was not meaningful and was not calculated. For those that maintained positive earnings, the ratio fell.

Source: *Standard and Poors' Compustat.*

Concept Check

9. How can a financial manager use the DuPont Identity to assess the firm's ROE?
10. How do you use the price-earnings (P/E) ratio to gauge the market value of a firm?

2.6 The Statement of Cash Flows

The income statement provides a measure of the firm's profit over a given time period. However, it does not indicate the amount of *cash* the firm has earned. There are two reasons that net income does not correspond to cash earned. First, there are non-cash entries on the income statement, such as depreciation and amortization. Second, certain uses, such as the purchase of a building or expenditures on inventory, and sources of cash, such as the collection of accounts receivable, are not reported on the income statement. The firm's **statement of cash flows** utilizes the information from the income statement and balance sheet to determine how much cash the firm has generated, and how that cash has been allocated, during a set period. Cash is important: It is needed to pay bills and maintain operations and is the source of any return of investment for investors. Thus, from the perspective of an investor attempting to value the firm or a financial manager concerned about cash flows (vs. earnings), the statement of cash flows provides what may be the most important information of the four financial statements.

statement of cash flows An accounting statement that shows how a firm has used the cash it earned during a set period.

The statement of cash flows is divided into three sections: operating activities, investment activities, and financing activities. These sections roughly correspond to the three major jobs of the financial manager.

1. *Operating activities* starts with net income from the income statement. It then adjusts this number by adding back all non-cash entries related to the firm's operating activities.
2. *Investment activities* lists the cash used for investment.
3. *Financing activities* shows the flow of cash between the firm and its investors.

Global's statement of cash flows is shown in Table 2.5. In this section, we take a close look at each component of the statement of cash flows.

Operating Activity

The first section of Global's statement of cash flows adjusts net income by all non-cash items related to operating activity. For instance, depreciation is deducted when computing net income, but it is not an actual cash expense. Thus, we add it back to net income when determining the amount of cash the firm has generated. Similarly, we add back any other non-cash expenses (for example, deferred taxes).

Next, we adjust for changes to net working capital that arise from changes to accounts receivable, accounts payable, or inventory. When a firm sells a product, it records the revenue as income even though it may not receive the cash from that sale immediately. Instead, it may grant the customer credit and let the customer pay in the future. The customer's obligation adds to the firm's accounts receivable. We use the following guidelines to adjust for changes in working capital:

1. Accounts receivable: When a sale is recorded as part of net income, but the cash has not yet been received from the customer, we must adjust the cash flows by *deducting* the increases in accounts receivable. This increase represents additional lending by the firm to its customers and it reduces the cash available to the firm.
2. Accounts payable: We *add* increases in accounts payable. Accounts payable represents borrowing by the firm from its suppliers. This borrowing increases the cash available to the firm.

TABLE 2.5

Global Corporation's Statement of Cash Flows for 2010 and 2009

GLOBAL CORPORATION
Statement of Cash Flows
Year ended December 31 (in $ millions)

	2010	2009
Operating activities		
Net income	2.0	1.9
Depreciation and amortization	1.2	1.1
Cash effect of changes in		
Accounts receivable	−5.3	−0.3
Accounts payable	2.7	−0.5
Inventory	−1.0	−1.0
Cash from operating activities	**−0.4**	**1.2**
Investment activities		
Capital expenditures	−33.4	−4.0
Acquisitions and other investing activity		
Cash from investing activities	**−33.4**	**−4.0**
Financing activities		
Dividends paid	−1.0	−1.0
Sale or purchase of stock	—	—
Increase in short-term borrowing	2.3	3.0
Increase in long-term borrowing	35.2	2.5
Cash from financing activities	**36.5**	**4.5**
Change in cash and cash equivalents	**2.7**	**1.7**

3. Inventory: Finally, we *deduct* increases in inventory. Increases in inventory are not recorded as an expense and do not contribute to net income (the cost of the goods are only included in net income when the goods are actually sold). However, the cost of increasing inventory is a cash expense for the firm and must be deducted.

Working capital adjustments address the difference between the time when sales and costs are recorded on the income statement and when the cash actually goes in and out of the firm. For example, Table 2.5 shows that in 2010 we *subtracted* the $5.3 million increase in accounts receivable from net income as part of the operating cash flow calculation. What happened? From Table 2.3, we see that Global had $10.6 million more in sales in 2010 than in 2009. However, from Table 2.1, we also see that Global's accounts receivable increased from $13.2 million in 2009 to $18.5 million in 2010. So, even though Global's sales were up considerably, it has not yet collected all the cash flow for those sales—instead, Global's customers owe it $5.3 million more at the end of 2010 than they did at the end of 2009. Because the statement of cash flows starts with net income, which includes sales for which Global has not yet been paid, we deduct the additional $5.3 million in sales Global is still owed when computing the actual cash flows it generated.

We must make a similar adjustment for inventory. Global does not record the cost of the inventory until it is sold, when it is included in the cost of goods sold. However, when it actually pays for the inventory, cash has flowed out of Global, decreasing operating cash flow. The opposite is true for accounts payable—Global has recorded additional expenses without actually paying for them yet. Those expenses reduce net income, but do not represent cash outflows.

Finally, we add depreciation to net income before calculating operating cash flow. Depreciation is an accounting adjustment to book value that is an expense, but not a cash outflow. That is, when Global's property, plant, and equipment depreciate by $1.2 million, it does not literally cost Global $1.2 million in cash flow. Because this is an expense that reduces net income, but not an actual cash outflow, we must add it back to calculate cash flow. We will talk more about depreciation when we do capital budgeting in Chapter 9. All these adjustments mean that cash flows can be very different from net income. Although Global showed positive net income on the income statement, it actually had a negative $0.4 million cash flow from operating activity, in large part because of the increase in accounts receivable.

Investment Activity

capital expenditures Purchases of new property, plant, and equipment.

The next section of the statement of cash flows shows the cash required for investment activities. Purchases of new property, plant, and equipment are **capital expenditures**. Recall that capital expenditures do not appear immediately as expenses on the income statement. Instead, the firm depreciates these assets and deducts depreciation expenses over time. To determine the firm's cash flow, we already added back depreciation because it is not an actual cash expense. Now, we subtract the actual capital expenditure that the firm made. Similarly, we also deduct other assets purchased or investments made by the firm, such as acquisitions. In Table 2.5, we see that in 2010, Global spent $33.4 million in cash on investing activities.

Financing Activity

The last section of the statement of cash flows shows the cash flows from financing activities. Dividends paid to shareholders are a cash outflow. Global paid $1 million to its shareholders as dividends in 2010.

retained earnings The difference between a firm's net income and the amount it spends on dividends.

payout ratio The ratio of a firm's dividends to its net income.

The difference between a firm's net income and the amount it spends on dividends is referred to as the firm's **retained earnings** for that year:

$$\text{Retained Earnings} = \text{Net Income} - \text{Dividends} \tag{2.20}$$

Global retained $2 million − $1 million = $1 million, or 50% of its earnings in 2010. This makes its *payout ratio* for 2010 equal to 50%. A firm's **payout ratio** is the ratio of its dividends to its net income:

$$\text{Payout Ratio} = \frac{\text{Dividends}}{\text{Net Income}} \tag{2.21}$$

Also listed under financing activity is any cash the company received from the sale of its own stock, or cash spent buying (repurchasing) its own stock. Global did not issue or repurchase stock during this period.

The last items to include in this section result from changes to Global's short-term and long-term borrowing. Global raised money by issuing debt, so the increases in short-term and long-term borrowing represent cash inflows. The last line of the statement of cash flows combines the cash flows from these three activities to calculate the overall change in the firm's cash balance over the period of the statement. In this case, Global had cash inflows of $2.7 million in 2010. By looking at the statement in Table 2.5 as a whole, we can determine that Global chose to borrow (mainly in the form of long-term debt) to cover the cost of its investment and operating activities. Although the firm's cash balance has increased, Global's negative operating cash flows and relatively high expenditures on investment activities might give investors some reasons for concern. If this pattern continues, Global will need to continue to borrow to remain in business.

EXAMPLE 2.5

The Impact of Depreciation on Cash Flow

Problem

Suppose Global had an additional $1 million depreciation expense in 2010. If Global's tax rate on pretax income is 26%, what would be the impact of this expense on Global's earnings? How would it impact Global's cash at the end of the year?

Solution

Plan

Depreciation is an operating expense, so Global's operating income, EBIT, and pretax income would be affected. With a tax rate of 26%, Global's tax bill will decrease by 26 cents for every dollar that pretax income is reduced. In order to determine how Global's cash would be impacted, we have to determine the effect of the additional depreciation on cash flows. Recall that depreciation is not an actual cash outflow, even though it is treated as an expense, so the only effect on cash flow is through the reduction in taxes.

Execute

Global's operating income, EBIT, and pretax income would fall by $1 million because of the $1 million in additional operating expense due to depreciation.

This $1 million decrease in pretax income would reduce Global's tax bill by $26\% \times \$1 \text{ million} = \$0.26 \text{ million}$. Therefore, net income would fall by $\$1 - \$0.26 = \$0.74$ million.

On the statement of cash flows, net income would fall by $0.74 million, but we would add back the additional depreciation of $1 million because it is not a cash expense. Thus, cash from operating activities would rise by $-\$0.74 + 1 = \$0.26$ million. Therefore, Global's cash balance at the end of the year would increase by $0.26 million, the amount of the tax savings that resulted from the additional depreciation deduction.

Evaluate

The increase in cash balance comes completely from the reduction in taxes. Because Global pays $0.26 million less in taxes even though its cash expenses have not increased, it has $0.26 million more in cash at the end of the year.

Concept Check

11. Why does a firm's net income not correspond to cash earned?
12. What are the components of the statement of cash flows?

2.7 Other Financial Statement Information

The most important elements of a firm's financial statements are the balance sheet, income statement, and the statement of cash flows, which we have already discussed. Several other pieces of information contained in the financial statements warrant brief mention: the *management discussion and analysis*, the *statement of stockholders' equity*, and *notes to the financial statement*.

management discussion and analysis (MD&A) A preface to the financial statements in which a company's management discusses the recent year (or quarter), providing a background on the company and any significant events that may have occurred.

off-balance sheet transactions Transactions or arrangements that can have a material impact on a firm's future performance yet do not appear on the balance sheet.

Management Discussion and Analysis

The **management discussion and analysis (MD&A)** is a preface to the financial statements in which the company's management discusses the recent year (or quarter), providing a background on the company and any significant events that may have occurred. Management may also discuss the coming year, and outline goals and new projects.

Management must also discuss any important risks that the firm faces or issues that may affect the firm's liquidity or resources. Management is also required to disclose any **off-balance sheet transactions**, which are transactions or arrangements that can have a material impact on the firm's future performance yet do not appear on the balance sheet. For example, if a firm has made guarantees that it will compensate a buyer for losses related to an asset purchased from the firm, these guarantees represent a potential future liability for the firm that must be disclosed as part of the MD&A.

would be affected and by how much. Also indicate the change to Global's book value of equity.

a. Global used $20 million of its available cash to repay $20 million of its long-term debt.
b. A warehouse fire destroyed $5 million worth of uninsured inventory.
c. Global used $5 million in cash and $5 million in new long-term debt to purchase a $10 million building.
d. A large customer owing $3 million for products it already received declared bankruptcy, leaving no possibility that Global would ever receive payment.
e. Global's engineers discover a new manufacturing process that will cut the cost of its flagship product by over 50%.
f. A key competitor announces a radical new pricing policy that will drastically undercut Global's prices.

5. What was the change in Global's book value of equity from 2009 to 2010 according to Table 2.1? Does this imply that the market price of Global's shares increased in 2010? Explain.
6. Use Google Finance (www.google.com/finance) to find the balance sheet data for Qualcomm as of the end of 2009.
   a. How much did Qualcomm have in cash and short-term investments?
   b. What were Qualcomm's total accounts receivable?
   c. What were Qualcomm's total assets?
   d. What were Qualcomm's total liabilities? How much of this was long-term debt?
   e. What was the book value of Qualcomm's equity?
7. Find the annual 10-K report for Peet's Coffee and Tea (PEET) online, filed for 2008 (filed in early 2009). Answer the following questions from its balance sheet:
   a. How much cash did Peet's have at the end of 2008?
   b. What were Peet's total assets?
   c. What were Peet's total liabilities? How much debt did Peet's have?
   d. What was the book value of Peet's equity?

Balance Sheet Analysis

8. In June 2007, General Electric (GE) had a book value of equity of $117 billion, 10.3 billion shares outstanding, and a market price of $38.00 per share. GE also had cash of $16 billion, and total debt of $467 billion.
   a. What was GE's market capitalization? What was GE's market-to-book ratio?
   b. What was GE's book debt-equity ratio? What was GE's market debt-equity ratio?
   c. What was GE's enterprise value?
9. In July 2007, Apple had cash of $7.12 billion, current assets of $18.75 billion, and current liabilities of $6.99 billion. It also had inventories of $0.25 billion.
   a. What was Apple's current ratio?
   b. What was Apple's quick ratio?
   c. In July 2007, Dell had a quick ratio of 1.25 and a current ratio of 1.30. What can you say about the asset liquidity of Apple relative to Dell?
10. In April 2010, the following information was true about Abercrombie and Fitch (ANF) and The Gap (GPS), both clothing retailers. Values (except price per share) are in millions of dollars.

	Book Equity	Price Per Share	Number of Shares
ANF	1,788	46.67	88.17
GPS	4,769	25.00	667.42

a. What is the market-to-book ratio of each company?
b. What conclusions do you draw from comparing the two ratios?

The Income Statement and Income Statement Analysis



11. Find online the annual 10-K report for Peet's Coffee and Tea (PEET) for 2008. Answer the following questions from the income statement:
a. What were Peet's revenues for 2008? By what percentage did revenues grow from 2007?
b. What were Peet's operating and net profit margins in 2008? How do they compare with its margins in 2007?
c. What were Peet's diluted earnings per share in 2008? What number of shares is this EPS based on?

12. Local Co. has sales of $10 million and cost of sales of $6 million. Its selling, general and administrative expenses are $500,000 and its research and development is $1 million. It has annual depreciation charges of $1 million and a tax rate of 35%.
a. What is Local's gross margin?
b. What is Local's operating margin?
c. What is Local's net profit margin?

13. If Local Co., the company in Problem 12, had an increase in selling expenses of $300,000, how would that affect each of its margins?

14. If Local Co., the company in Problem 12, had interest expense of $800,000, how would that affect each of its margins?

15. Chutes & Co. has interest expense of $1 million and an operating margin of 10% on total sales of $30 million. What is Chutes' interest coverage ratio?

16. Ladders, Inc. has a net profit margin of 5% on sales of $50 million. It has book value of equity of $40 million and total liabilities with a book value of $30 million. What is Ladders' ROE? ROA?

17. JPJ Corp has sales of $1 million, accounts receivable of $50,000, total assets of $5 million (of which $3 million are fixed assets), inventory of $150,000, and cost of goods sold of $600,000. What is JPJ's accounts receivable days? Fixed asset turnover? Total asset turnover? Inventory turnover?

18. If JPJ Corp (the company from the previous question) is able to increase sales by 10% but keep its total and fixed asset growth to only 5%, what will its new asset turnover ratios be?



***19.** Suppose that in 2010, Global launched an aggressive marketing campaign that boosted sales by 15%. However, their operating margin fell from 5.57% to 4.50%. Suppose that they had no other income, interest expenses were unchanged, and taxes were the same percentage of pretax income as in 2009.
a. What was Global's EBIT in 2010?
b. What was Global's income in 2010?
c. If Global's P/E ratio and number of shares outstanding remained unchanged, what was Global's share price in 2010?

20. Suppose a firm's tax rate is 35%.
a. What effect would a $10 million operating expense have on this year's earnings? What effect would it have on next year's earnings?
b. What effect would a $10 million capital expense have on this year's earnings, if the capital is depreciated at a rate of $2 million per year for five years? What effect would it have on next year's earnings?

21. You are analyzing the leverage of two firms and you note the following (all values in millions of dollars):

	Debt	Book Equity	Market Equity	Operating Income	Interest Expense
Firm A	500	300	400	100	50
Firm B	80	35	40	8	7

a. What is the market debt-to-equity ratio of each firm?
b. What is the book debt-to-equity ratio of each firm?
c. What is the interest coverage ratio of each firm?
d. Which firm will have more difficulty meeting its debt obligations?

22. For 2010, Wal-Mart and Target had the following information (all values are in millions of dollars):

	Sales (Income Statement)	Cost of Goods Sold (Income Statement)	Accounts receivable (Balance Sheet)	Inventory (Balance Sheet)
Wal-Mart	408,214	304,657	4,144	30,254
Target	65,357	45,583	6,966	7,179

a. What is each company's accounts receivable days?
b. What is each company's inventory turnover?
c. Which company is managing its accounts receivable and inventory more efficiently?

***23.** Quisco Systems has 6.5 billion shares outstanding and a share price of $18.00. Quisco is considering developing a new networking product in-house at a cost of $500 million. Alternatively, Quisco can acquire a firm that already has the technology for $900 million worth (at the current price) of Quisco stock. Suppose that absent the expense of the new technology, Quisco will have EPS of $0.80.
a. Suppose Quisco develops the product in house. What impact would the development cost have on Quisco's EPS? Assume all costs are incurred this year and are treated as an R&D expense, Quisco's tax rate is 35%, and the number of shares outstanding is unchanged.
b. Suppose Quisco does not develop the product in house but instead acquires the technology. What effect would the acquisition have on Quisco's EPS this year? (Note that acquisition expenses do not appear directly on the income statement. Assume the acquired firm has no revenues or expenses of its own, so that the only effect on EPS is due to the change in the number of shares outstanding.)
c. Which method of acquiring the technology has a smaller impact on earnings? Is this method cheaper? Explain.

24. In January 2009, American Airlines (AMR) had a market capitalization of $1.7 billion, debt of $11.1 billion, and cash of $4.6 billion. American Airlines had revenues of $23.8 billion. British Airways (BABWF) had a market capitalization of $2.2 billion, debt of $4.7 billion, cash of $2.6 billion, and revenues of $13.1 billion.
a. Compare the market capitalization-to-revenue ratio (also called the price-to-sales ratio) for American Airlines and British Airways.
b. Compare the enterprise value-to-revenue ratio for American Airlines and British Airways.
c. Which of these comparisons is more meaningful? Explain.

***25.** Find online the annual 10-K report for Peet's Coffee and Tea (PEET) for 2008 (filed in early 2009).
a. Compute Peet's net profit margin, total asset turnover, and equity multiplier.
b. Verify the DuPont Identity for Peet's ROE.
c. If Peet's managers wanted to increase its ROE by 1 percentage point, how much higher would their asset turnover need to be?

26. Repeat the analysis from parts a and b of the previous problem using Starbucks Corporation (SBUX) instead. Based on the DuPont Identity, what explains the difference between the two firms' ROEs?

27. Consider a retail firm with a net profit margin of 3.5%, a total asset turnover of 1.8, total assets of $44 million, and a book value of equity of $18 million.
a. What is the firm's current ROE?
b. If the firm increased its net profit margin to 4%, what would its ROE be?
c. If, in addition, the firm increased its revenues by 20% (while maintaining this higher profit margin and without changing its assets or liabilities), what would its ROE be?

The Statement of Cash Flows

28. Find online the 2008 annual 10-K report for Peet's Coffee and Tea (PEET), filed in early 2009. Answer the following questions from its cash flow statement:
a. How much cash did Peet's generate from operating activities in 2008?
b. What was Peet's depreciation expense in 2008?
c. How much cash was invested in new property and equipment (net of any sales) in 2008?
d. How much did Peet's raise from the sale of shares of its stock (net of any purchases) in 2008?



29. See the cash flow statement below for H. J. Heinz (HNZ) (all values in thousands of dollars) (*see MyFinanceLab for the data in Excel format*):
a. What were Heinz's cumulative earnings over these four quarters? What were its cumulative cash flows from operating activities?

Period Ending	29-Oct-08	30-Jul-08	30-Apr-08	30-Jan-08
Net income	**276,710**	**228,964**	**194,062**	**218,532**
Operating Activities, Cash Flows Provided By or Used In				
Depreciation	69,997	75,733	74,570	73,173
Adjustments to net income	14,359	(13,142)	48,826	(47,993)
Changes In accounts receivables	(38,869)	(53,218)	100,732	(84,711)
Changes in liabilities	82,816	(111,577)	201,725	39,949
Changes in inventories	(195,186)	(114,121)	85,028	57,681
Changes in other operating activities	17,675	(26,574)	12,692	(2,097)
Total Cash Flow From Operating Activities	**227,502**	**(13,935)**	**717,635**	**254,534**
Investing Activities, Cash Flows Provided By or Used In				
Capital expenditures	(82,584)	(41,634)	(100,109)	(69,170)
Investments	(5,465)	5,465	(93,153)	(48,330)
Other cashflows from investing activities	(108,903)	732	(58,069)	20,652
Total Cash Flows From Investing Activities	**(196,952)**	**(35,437)**	**(251,331)**	**(96,848)**
Financing Activities, Cash Flows Provided By or Used In				
Dividends paid	(131,483)	(131,333)	(119,452)	(121,404)
Sale purchase of stock	78,774	1,210	(76,807)	(79,288)
Net borrowings	515,709	114,766	(283,696)	64,885
Other cash flows from financing activities	(282)	2,000	(46,234)	39,763
Total Cash Flows From Financing Activities	**462,718**	**(13,357)**	**(526,189)**	**(96,044)**
Effect of exchange rate changes	(119,960)	(610)	32,807	6,890
Change In Cash and Cash Equivalents	**$373,308**	**(63,339)**	**(27,078)**	**$68,532**

b. What fraction of the cash from operating activities was used for investment over the four quarters?
c. What fraction of the cash from operating activities was used for financing activities over the four quarters?

30. Suppose your firm receives a $5 million order on the last day of the year. You fill the order with $2 million worth of inventory. The customer picks up the entire order the same day and pays $1 million up front in cash; you also issue a bill for the customer to pay the remaining balance of $4 million within 40 days. Suppose your firm's tax rate is 0% (i.e., ignore taxes). Determine the consequences of this transaction for each of the following:

a. Revenues
b. Earnings
c. Receivables
d. Inventory
e. Cash

31. Nokela Industries purchases a $40 million cyclo-converter. The cyclo-converter will be depreciated by $10 million per year over four years, starting this year. Suppose Nokela's tax rate is 40%.

a. What impact will the cost of the purchase have on earnings for each of the next four years?
b. What impact will the cost of the purchase have on the firm's cash flow for the next four years?

Other Financial Statement Information

32. The balance sheet information for Clorox Co. (CLX) in 2004–2005 is shown here (all values in thousands of dollars) (*see MyFinanceLab for the data in Excel format*):

Balance Sheet:	31-Mar-05	31-Dec-04	30-Sep-04	30-Jun-04
Assets				
Current Assets				
Cash and cash equivalents	293,000	300,000	255,000	232,000
Net receivables	401,000	362,000	385,000	460,000
Inventory	374,000	342,000	437,000	306,000
Other current assets	60,000	43,000	53,000	45,000
Total Current Assets	**1,128,000**	**1,047,000**	**1,130,000**	**1,043,000**
Long-term investments	128,000	97,000	—	200,000
Property, plant, and equipment	979,000	991,000	995,000	1,052,000
Goodwill	744,000	748,000	736,000	742,000
Other assets	777,000	827,000	911,000	797,000
Total Assets	**3,756,000**	**3,710,000**	**3,772,000**	**3,834,000**
Liabilities				
Current Liabilities				
Accounts payable	876,000	1,467,000	922,000	980,000
Short/current long-term debt	410,000	2,000	173,000	288,000
Other current liabilities	—	—	—	—
Total Current Liabilities	**1,286,000**	**1,469,000**	**1,095,000**	**1,268,000**
Long-term debt	2,381,000	2,124,000	474,000	475,000
Other liabilities	435,000	574,000	559,000	551,000
Total Liabilities	**4,102,000**	**4,167,000**	**2,128,000**	**2,294,000**
Total Stockholder Equity	**–346,000**	**–457,000**	**1,644,000**	**1,540,000**
Total Liabilities and Stockholder Equity	**$3,756,000**	**$3,710,000**	**$3,772,000**	**$3,834,000**

Ross

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- Fluxo de caixa: FCO, gastos líquidos de capital, variação do CCL, credores e acionistas: Cap. 2, §2.4 (p. 3

pais. E há ainda contribuições compulsórias para entidades de classe como Senai, Senac, Sesi, Senar e sindicatos. Todos esses tributos são repassados para o preço, ou seja, é o consumidor quem paga. É verdade que o consumidor também paga o imposto de renda das empresas, pois afinal, ao calcular as suas margens, as empresas já incluem um cálculo para o imposto de renda e contribuição social sobre os lucros. Portanto, pense duas vezes antes de propor aumento de impostos para empresas: é você quem vai pagar! Os impostos sobre o faturamento passam a existir pelo simples fato de uma nota fiscal ser extraída, de forma que a empresa que vende é meramente um caixa do erário público nas diferentes instâncias: federal, estadual e municipal. Ela cobra os tributos e os repassa para o erário. O consumidor vê preço e as empresas veem tributos. Os tributos sobre a folha de pagamento incluem as contribuições para INSS e FGTS e programas de formação de patrimônio do trabalhador. Estes tributos existem sempre que uma empresa empregar alguém.

O imposto de renda e a contribuição social no regime do lucro real são impostos contingentes, só existem se lucro houver e esse lucro pode ser modificado pelas despesas financeiras. É por isso que a consideração dos impostos sobre os lucros da pessoa jurídica é importante para o administrador financeiro. O imposto de renda da pessoa jurídica (IRPJ) é um dos fatores de decisão na escolha da estrutura de capital e do custo de capital. Veremos isso especialmente nos Capítulos 15 e 16. Por ora nossa recomendação é: sempre simule tributos nos diferentes regimes, pois algo que parece mais fácil pode lhe custar mais caro.

2.4 Fluxo de caixa

Neste ponto, estamos prontos para discutir uma das informações financeiras talvez mais importantes que podem ser garimpadas nas demonstrações financeiras: o fluxo de caixa. *Fluxo de caixa* é a diferença entre a quantidade de dinheiro que entrou e a quantidade que saiu. Por exemplo, se você fosse o proprietário de um negócio, poderia estar muito interessado na quantidade de caixa que realmente tirou da sua empresa em determinado ano. Como determinar essa quantidade é uma das coisas que discutiremos a seguir.

Nenhuma demonstração contábil padronizada apresenta essas informações da forma como desejaríamos. Por isso, discutiremos como calcular o fluxo de caixa da Brasil S/A e destacaremos a diferença entre esse resultado e os cálculos da demonstração contábil padronizada. Existe uma demonstração contábil chamada *demonstração de fluxos de caixa*, mas ela diz respeito a uma questão um pouco diferente, e não deve ser confundida com aquilo que é discutido nesta seção. A demonstração contábil dos fluxos de caixa será discutida no Capítulo 3.

A partir da identidade do balanço patrimonial, sabemos que o valor do ativo de uma empresa é igual ao valor de seu passivo mais o valor do patrimônio dos acionistas. Da mesma forma, o fluxo de caixa dos ativos da empresa deve ser igual à soma do fluxo de caixa para os credores e do fluxo de caixa para os acionistas (ou proprietários):

$$\begin{aligned}\text{Fluxo de caixa dos ativos} = {} & \text{Fluxo de caixa para os credores} + \\ & \text{Fluxo de caixa para os acionistas}\end{aligned} \qquad [2.3]$$

Essa é a *identidade do fluxo de caixa*. Ela diz que o fluxo de caixa dos ativos da empresa é igual ao fluxo de caixa pago aos fornecedores de capital da empresa (acionistas e credores). Ela reflete o fato de que uma empresa gera caixa por meio de suas diversas atividades e que o caixa é usado para pagar credores e proprietários. A seguir, discutiremos os diversos itens que formam esses fluxos de caixa.

fluxo de caixa dos ativos
é a soma do fluxo de caixa operacional, dos gastos de capital e da variação do capital circulante líquido.

fluxo de caixa operacional
o caixa gerado pelas atividades operacionais normais de uma empresa.

Fluxo de caixa dos ativos

O **fluxo de caixa dos ativos** envolve três componentes: o fluxo de caixa operacional, os gastos de capital e a variação do capital circulante líquido. O **fluxo de caixa operacional** refere-se ao fluxo de caixa resultante das atividades diárias de produção e vendas. As despesas

associadas ao financiamento dos ativos da empresa não estão incluídas porque não são despesas operacionais[†].

Como discutimos no Capítulo 1, parte do fluxo de caixa é reinvestido na empresa. Os gastos de capital se referem aos gastos líquidos com ativos não circulantes (as compras de ativos não circulantes menos as vendas de ativos não circulantes). Finalmente, a *variação do capital circulante líquido* é medida como a variação líquida do ativo circulante em relação ao passivo circulante no período examinado e representa a quantidade gasta em capital circulante líquido. Os três componentes do fluxo de caixa serão examinados em mais detalhe a seguir.

Fluxo de caixa operacional Para calcular o fluxo de caixa operacional (FCO), devemos calcular as receitas menos os custos, mas não deduzir a depreciação, porque ela não é uma saída de caixa, nem incluir juros, porque são uma despesa de financiamento. Incluiremos os impostos, porque eles (infelizmente) são pagos à vista.

Se examinarmos a demonstração de resultados da Brasil S/A (Tabela 2.2), veremos que os lucros antes de juros e impostos (Lajir) são de $694. Isso é quase o que queremos, pois não inclui os juros pagos. Precisamos fazer dois ajustes. Em primeiro lugar, lembre-se de que a depreciação é uma despesa que não afeta o caixa. Para obter o fluxo de caixa, primeiro somamos novamente os $65 de depreciação, porque essa não foi uma saída de caixa. O outro ajuste é subtrair os $212 de impostos, porque eles foram pagos à vista. O resultado é o fluxo de caixa operacional:

BRASIL S/A Fluxo de caixa operacional em 2009	
Lucros antes de juros e impostos	$694
+ Depreciação	65
− Impostos	212
Fluxo de caixa operacional	$547

Assim, a Brasil S/A teve um fluxo de caixa operacional de $547 em 2009.

O fluxo de caixa operacional é um número importante porque nos diz, em um nível muito básico, se os fluxos de entrada de caixa das operações comerciais de uma empresa são ou não suficientes para cobrir seus fluxos diários de saídas de caixa. Por esse motivo, um fluxo de caixa operacional negativo quase sempre indica problemas.

Existe uma desagradável possibilidade de confusão quando falamos do fluxo de caixa operacional. Na prática contábil, o fluxo de caixa operacional quase sempre é definido como lucro líquido mais depreciação. Para a Brasil S/A, isso seria $412 + $65 = $477.

A definição contábil de fluxo de caixa operacional tem uma diferença importante em relação à nossa: os juros são deduzidos quando o lucro líquido é calculado. Observe que a diferença entre o fluxo de caixa operacional de $547 que calculamos e esses $477 é de $70, o montante de juros pagos no ano. Essa definição de fluxo de caixa, portanto, considera os juros pagos como despesa operacional. Nossa definição trata como despesa financeira. Se não houvesse despesas de juros, as duas definições seriam iguais.

Para encerrar nosso cálculo do fluxo de caixa dos ativos da Brasil S/A, precisamos levar em conta quanto do fluxo de caixa operacional de $547 foi reinvestido na empresa. Consideramos primeiro as despesas com ativos imobilizados.

Gastos de capital Os gastos líquidos de capital são apenas o valor gasto com ativos imobilizados menos o valor recebido da venda de ativos imobilizados. Ao final de 2008, o ativo imobilizado líquido da Brasil S/A (Tabela 2.1) foi de $1.644. Durante o ano, ela deu baixa

[†] N. de R. T.: Em algumas circunstâncias, custos de empréstimos que são diretamente atribuíveis à aquisição, à construção ou à produção de ativo que demanda um período de tempo substancial para ficar pronto para seu uso ou venda pretendidos formam parte do custo de tal ativo, e nesse caso são gastos de capital. Outros custos de empréstimos devem ser reconhecidos como despesas. Ver pronunciamento CPC 20 – Empréstimos no *site* do Comitê de Pronunciamentos Contábeis.

(depreciou) de $65 em ativos imobilizados na demonstração de resultados do exercício. Dessa forma, se a empresa não comprou ativo imobilizado, o ativo imobilizado líquido teria sido de $1.644 − $65 = $1.579 ao final do ano. O balanço patrimonial de 2009 mostra $1.709 em ativo imobilizado líquido, de modo que a Brasil S/A deve ter gasto um total de $1.709 − $1.579 = $130 em ativo imobilizado durante o ano:

Ativo imobilizado líquido final	$1.709
− Ativo imobilizado líquido inicial	1.644
+ Depreciação	65
Gastos líquidos de capital	$ 130

Esses $130 são os gastos líquidos de capital em 2009.

Os gastos líquidos de capital poderiam ser negativos? A resposta é sim. Isso aconteceria se a empresa tivesse vendido mais ativos do que comprado. O *líquido* aqui se refere a compras de ativo imobilizado menos vendas de ativo imobilizado. Com frequência, você verá os gastos de capital sendo chamados de Capex (abreviação de *capital expenditures*).

Variação do capital circulante líquido Além de investir em ativo imobilizado, uma empresa também investirá em ativo circulante. Por exemplo, voltando aos balanços patrimoniais da Tabela 2.1, vemos que, ao final de 2009, a Brasil S/A tinha um ativo circulante de $1.403. Ao final de 2008, o ativo circulante era de $1.112 e, portanto, durante o ano, a empresa investiu $1.403 − $1.112 = $291 em ativos circulantes.

Conforme a empresa muda seus investimentos em ativos circulantes, em geral, seu passivo circulante também mudará. Para determinar a variação do capital circulante líquido, a abordagem mais simples é assumir a diferença entre o capital circulante líquido (CCL) inicial e final. O capital circulante líquido, ao final de 2009, foi de $1.403 − $389 = $1.014. Da mesma forma, ao final de 2008, o capital circulante líquido foi de $1.112 − $428 = $684. Dados esses números, temos o seguinte:

CCL final	$1.014
− CCL inicial	684
Variação do CCL	$ 330

Portanto, o capital circulante líquido aumentou em $330. Em outras palavras, a Brasil S/A teve um investimento líquido de $330 em CCL no ano. Essa variação do CCL com frequência é chamada de "variação do circulante".

Conclusão Com os números que obtivemos, estamos prontos para calcular o fluxo de caixa dos ativos. O fluxo de caixa total dos ativos é o fluxo de caixa operacional menos os montantes investidos em ativos imobilizados e em capital circulante líquido. Assim, para a Brasil S/A temos:

BRASIL S/A **Fluxo de caixa dos ativos em 2009**	
Fluxo de caixa operacional	$547
− Gastos líquidos de capital	130
− Variação do CCL	330
Fluxo de caixa dos ativos	$ 87

Pela identidade do fluxo de caixa vista anteriormente, sabemos que esse fluxo de caixa dos ativos de $87 é igual à soma do fluxo de caixa para os credores e do fluxo de caixa para os acionistas. Veremos isso a seguir.

Um fluxo de caixa negativo é bastante comum para uma empresa em crescimento. Como veremos a seguir, um fluxo de caixa negativo significa que a empresa levantou mais dinheiro com empréstimos e com a emissão de novas ações do que pagou para os credores e os acionistas durante o ano.

fluxo de caixa livre
outro nome para fluxo de caixa dos ativos.

Observação sobre fluxo de caixa "livre" Às vezes, o fluxo de caixa dos ativos recebe um nome diferente: **fluxo de caixa livre**. Obviamente, não existe dinheiro "livre", sem dono (quem dera!). Em vez disso, o nome se refere ao caixa que a empresa pode distribuir livremente aos credores e aos acionistas porque não é necessário para o capital de giro ou para investimentos em novos ativos imobilizados. Permaneceremos com "fluxo de caixa dos ativos" como rótulo para esse importante conceito, porque na prática existe certa variação na forma como o fluxo de caixa livre é calculado; usuários diferentes o calculam de maneiras diferentes. No entanto, sempre que ouvir "fluxo de caixa livre", entenda que a discussão se refere ao fluxo de caixa dos ativos ou algo muito parecido com isso.

fluxo de caixa para os credores
pagamentos de juros de uma empresa para os credores menos os novos empréstimos líquidos.

fluxo de caixa para acionistas
dividendos e juros sobre o capital próprio pagos por uma empresa menos o capital novo obtido.

Fluxo de caixa para credores e acionistas

Os fluxos de caixa para credores e acionistas representam os pagamentos líquidos a credores e proprietários durante o ano. Seu cálculo é semelhante ao cálculo do fluxo de caixa dos ativos. **Fluxo de caixa para os credores** são os juros pagos menos os novos empréstimos líquidos, e **fluxo de caixa para os acionistas** são os dividendos e juros sobre o capital próprio pagos aos acionistas menos os aumentos de capital realizados pelos acionistas.

Fluxo de caixa para os credores Examinando a demonstração de resultados da Tabela 2.2, vemos que a Brasil S/A pagou $70 de juros para os credores. Pelos balanços patrimoniais da Tabela 2.1, observamos que o passivo não circulante aumentou em $454 − $408 = $46. Assim, a empresa pagou $70 de juros, mas tomou mais $46 emprestados. O fluxo de caixa líquido para os credores, portanto, é:

BRASIL S/A Fluxo de caixa para os credores em 2009	
Juros pagos	$70
− Novos empréstimos (líquido)	46
Fluxo de caixa para credores	$24

O fluxo de caixa para os credores, às vezes, é chamado de *fluxo de caixa para detentores de títulos de dívida*; usaremos ambos os termos.

Fluxo de caixa para os acionistas Na demonstração de resultados, vemos que os dividendos pagos aos acionistas somaram $103. Para obter o valor do novo aporte de capital, precisamos examinar o capital social e ágio recebido. Essa conta nos diz o quanto a empresa obteve com a venda de novas ações. Durante o ano, essa conta aumentou em $40, portanto, o aumento de capital foi $40. Assim, temos:

BRASIL S/A Fluxo de caixa para acionistas em 2009	
Dividendos pagos	$103
− Aumento de capital	40
Fluxo de caixa para acionistas	$ 63

Portanto, o fluxo de caixa para os acionistas foi de $63 em 2009.

A última coisa é verificar se a identidade do fluxo de caixa se mantém, para ter certeza de que não cometemos erro algum. Conforme vimos na seção anterior, o fluxo de caixa dos ativos é $87. O fluxo de caixa para credores e acionistas é de $24 + $63 = $87 e, portanto, tudo confere. A Tabela 2.5 contém um resumo dos diversos cálculos do fluxo de caixa para referência futura.

Como indica nossa discussão, é essencial que uma empresa controle seu fluxo de caixa. O texto a seguir é um excelente lembrete de que isso é uma boa ideia, a menos que os proprietários queiram acabar na rua da amargura.

TABELA 2.5 Resumo do fluxo de caixa

I. A identidade do fluxo de caixa	
Fluxo de caixa dos ativos	= Fluxo de caixa para os credores (detentores de títulos de dívida) + Fluxo de caixa para os acionistas (proprietários)
II. Fluxo de caixa dos ativos	
Fluxo de caixa dos ativos	= Fluxo de caixa operacional − Despesas de capital líquidas − Variação do capital circulante líquido (CCL)
onde:	
Fluxo de caixa operacional	= Lucros antes de juros e impostos (Lajir) + Depreciação − Impostos
Gastos líquidos de capital	= Ativo imobilizado líquido final − Ativo imobilizado líquido inicial + Depreciação
Variação do CCL	= CCL final − CCL inicial
III. Fluxo de caixa para os credores (detentores de títulos de dívida)	
Fluxo de caixa para os credores	= Juros pagos − Novos empréstimos líquidos
IV. Fluxo de caixa para os acionistas (proprietários)	
Fluxo de caixa para os acionistas	= Dividendos e juros sobre o capital próprio pagos − Aumento de capital

ASSIM DISSE O BANQUEIRO: "CUIDADO COM O FLUXO DE CAIXA"

Em uma noite sombria, enquanto meditava fraco e fatigado
Sobre um curioso livro de conhecimento contábil,
Buscando algum truque (inescrupuloso) para fugir por entre os dedos fiscais,
De repente, ouvi uma batida à minha porta,
Apenas isso, e nada mais.

Em seguida, senti um calafrio desagradável e ouvi o dinheiro tilintar
Quando um banqueiro assustador, que já havia visto antes, se insinuou a entrar.
Seu rosto tinha os tons do dinheiro e em seus olhos eu podia ver
Cifrões que brilhavam, enquanto aquela dívida ele não parava de calcular.
"Fluxo de caixa", disse o banqueiro, e nada mais.

Eu sempre achei interessante mostrar um lucro em azul brilhante.
Mas o banqueiro ecoou um sonoro "Não.
As contas a receber têm tamanho acúmulo, que só de olhar penso em seu túmulo;
Avolumam-se as baixas. O que importa é o fluxo de caixa."
Ele repetiu, "Cuidado com o fluxo de caixa".

Eu tentei contar sem choque a história de nosso belo estoque
Que até é grande, mas tem muita coisa bem-apanhada.
Mas o banqueiro viu seu crescimento e, praguejando terrulento,
Sacudiu os braços e gritou, "Pare! Não diga mais nada!
Pague os juros, e chega de conversa fiada!"

A seguir, procurei itens não caixa que pudessem ser adicionados *ad infinitum*
Para repor a saída de caixa que ocorria sem cessar,
Mas, para tirar a demonstração do vermelho, segurei a depreciação sem conselho,
E meu banqueiro disse que eu tinha feito algo sem noção.
Ele estremeceu, e o ranger dos dentes denotava indignação.

Quando pedi por financiamento, ele respondeu com um lamento
Que a taxa avarenta seria o CDI mais cinquenta,
E, para garantir minha boa valia, ele insistiu que houvesse garantia —
Todo o meu ativo mais o couro da minha mão.
Apenas isso, uma taxa padrão.

Embora eu esteja fora do vermelho, estou apanhando de relho,
Meu caixa só tem saídas e os clientes pagam devagar.
Já há muitas contas a receber e elas não param de crescer:
O resultado é certo – uma aflição de matar!
E eu ouço o banqueiro produzir um gemido feito horrendamente para desiludir,
"Cuidado com o fluxo de caixa".

Herbert S. Bailey Jr.

Fonte: Reimpresso da edição de 13 de janeiro de 1975, da *Publishers Weekly*, publicada pela R. R. Bowker, uma empresa do grupo Xerox. Copyright © 1975 de Xerox Corporation.

Um exemplo: fluxos de caixa da Esmola Cola

O exemplo a seguir, mais detalhado, aborda os diversos cálculos do fluxo de caixa discutidos neste capítulo. Ele também ilustra algumas variações que podem surgir.

Fluxo de caixa operacional Durante o ano, a Esmola Cola S/A teve vendas e custo das mercadorias vendidas de $600 e $300, respectivamente. A depreciação foi de $150 e os juros pagos foram de $30. Os impostos foram calculados a 34%. Os dividendos foram de $30. (Todos os valores são em milhões.) Qual foi o fluxo de caixa operacional da Esmola? Por que ele é diferente do lucro líquido?

O mais fácil a fazer neste caso é ir em frente e criar uma demonstração de resultados do exercício. Em seguida, podemos escolher os números necessários. A demonstração de resultados da Esmola Cola é esta:

ESMOLA COLA Demonstração de resultados de 2009		
Vendas líquidas		$600
Custo das mercadorias vendidas		300
Depreciação		150
Lucro antes dos juros e dos impostos		$150
Juros pagos		30
Lucro tributável		$120
Impostos		41
Lucro líquido		$ 79
Dividendos	$30	
Acréscimo aos lucros retidos	49	

Assim, o lucro líquido da Esmola totalizou $79. Agora temos todos os números de que precisamos. Tomando como referência o exemplo da Brasil S/A e a Tabela 2.5, temos:

ESMOLA COLA Fluxo de caixa operacional em 2009	
Lucro antes de juros e impostos	$150
+ Depreciação	150
− Impostos	41
Fluxo de caixa operacional	$259

Como esse exemplo ilustra, fluxo de caixa operacional não é o mesmo que lucro líquido porque, no cálculo deste, são subtraídos os juros e a depreciação. Na discussão anterior, não subtraímos esses itens no cálculo do fluxo de caixa operacional porque a depreciação não é uma despesa de caixa, e consideramos os juros pagos uma despesa financeira, e não operacional.

Gastos líquidos de capital Suponhamos que o ativo imobilizado líquido inicial fosse $500 e o ativo imobilizado líquido final, $750. Qual seria o gasto líquido de capital no ano?

Pela demonstração financeira da Esmola, sabemos que a depreciação do ano foi de $150. O ativo imobilizado líquido aumentou em $250. Portanto, a empresa gastou $250 mais $150, totalizando $400.

Variação do CCL e do fluxo de caixa dos ativos Suponhamos que a Esmola Cola tenha iniciado o ano com $2.130 em ativo circulante e $1.620 em passivo circulante, e que os números finais correspondentes fossem $2.260 e $1.710. Qual foi a variação do CCL durante o ano? Qual foi o fluxo de caixa dos ativos? Como isso pode ser comparado ao lucro líquido?

O capital circulante líquido começou em $2.130 − $1.620 = $510 e terminou em $2.260 − $1.710 = $550. O acréscimo ao CCL, portanto, foi de $550 − $510 = $40. Juntando todas as informações da Esmola, temos:

ESMOLA COLA Fluxo de caixa dos ativos em 2009	
Fluxo de caixa operacional	$259
− Gastos líquidos de capital	400
− Variação do CCL	40
Fluxo de caixa dos ativos	− $181

A Esmola teve um fluxo de caixa dos ativos de – $181. O lucro líquido estava positivo em $79. O fato de o fluxo de caixa dos ativos ter sido negativo é preocupante? Não necessariamente. O fluxo de caixa aqui é negativo principalmente por causa de um grande investimento em ativo imobilizado. Se foi um bom investimento, o resultado negativo não é preocupante.

Fluxo de caixa para acionistas e credores Dissemos que a Esmola Cola teve fluxo de caixa do ativo de − $181. O fato de ser negativo significa que, no ano, a Esmola levantou mais dinheiro com novas dívidas e com novos aportes de capital do que pagou. Por exemplo, suponhamos que ela não vendeu ações novas no ano. Qual foi o fluxo de caixa para os acionistas? E para os credores?

Como a Esmola não levantou capital novo, seu fluxo de caixa para os acionistas é igual aos dividendos pagos:

ESMOLA COLA Fluxo de caixa para os acionistas em 2009	
Dividendos pagos	$30
− Capital novo	0
Fluxo de caixa para os acionistas	$30

Agora, da identidade dos fluxos de caixa sabemos que o total pago aos credores e aos acionistas foi de − $181. O fluxo de caixa para os acionistas foi de $30, então o fluxo de caixa para os credores tem de ser igual a − $181 − $30 = − $211:

$$\begin{aligned} &\text{Fluxo de caixa para os credores} + \text{Fluxo de caixa para os acionistas} = -\$181 \\ &\text{Fluxo de caixa para os credores} + \$30 = -\$181 \\ &\text{Fluxo de caixa para os credores} = -\$211 \end{aligned}$$

Sabendo que o fluxo de caixa para os credores é de − $211 e que os juros pagos totalizam $30, agora podemos determinar o valor líquido dos novos empréstimos. A Esmola deve ter tomado emprestado $241 durante o ano, para ajudar a financiar a expansão do ativo imobilizado:

ESMOLA COLA Fluxo de caixa para os credores em 2009	
Juros pagos	$30
− Valor líquido dos empréstimos novos	− 241
Fluxo de caixa para os credores	−$211

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Questões conceituais

2.4a O que é a identidade do fluxo de caixa? Explique o que significa.

2.4b Quais são os componentes do fluxo de caixa operacional?

2.4c Por que os juros pagos não são um componente do fluxo de caixa operacional?

2.5 Resumo e conclusões

Este capítulo apresentou alguns fundamentos das demonstrações financeiras, dos impostos e do fluxo de caixa. Vimos que:

1. Os valores contábeis de um balanço patrimonial podem ser muito diferentes dos valores de mercado. E o objetivo da administração financeira é maximizar o valor de mercado das ações, não seu valor contábil.
2. O lucro líquido calculado na demonstração de resultados não é igual ao fluxo de caixa. Um dos principais motivos para isso é que a depreciação, uma despesa que não afeta o caixa, é deduzida como despesa quando do cálculo do lucro líquido.
3. As alíquotas tributárias médias e marginais podem ser diferentes, e a alíquota tributária marginal é o item relevante para a maioria das decisões financeiras.
4. A alíquota tributária marginal paga pelas empresas brasileiras com as maiores receitas é de 34%.
5. Assim como há uma identidade para o balanço patrimonial, há uma identidade para o fluxo de caixa. Ela mostra que o fluxo de caixa dos ativos é igual à soma dos fluxos de caixa para credores e acionistas.

O cálculo do fluxo de caixa das demonstrações financeiras não é difícil. Mas é preciso tomar cuidado ao lidar com despesas que não afetam o caixa, tais como a depreciação, e não confundir despesas operacionais com despesas de financiamento. Acima de tudo, é importante não confundir valores contábeis com valores de mercado, ou receita contábil com fluxo de caixa.

REVISÃO DO CAPÍTULO E TESTE DE CONHECIMENTOS

2.1 Fluxo de caixa da Corporação Mara Este problema permite praticar com as demonstrações financeiras e descobrir o fluxo de caixa. Com base nas informações a seguir sobre a Corporação Mara, monte uma demonstração de resultados de 2009 e os balanços patrimoniais de 2008 e 2009. Em seguida, tomando como base os exemplos da Brasil S/A mostrados neste capítulo, calcule o fluxo de caixa dos ativos, o fluxo de caixa para os credores e o fluxo de caixa para os acionistas da Mara em 2009. Use uma alíquota de 34%. Você pode conferir suas respostas com as nossas, que se encontram na próxima seção.

	2008	2009
Vendas	$4.203	$4.507
Custo de mercadorias vendidas	2.422	2.633
Depreciação	785	952
Juros	180	196
Dividendos	225	250
Ativo circulante	2.205	2.429
Ativo não circulante líquido	7.344	7.650
Passivo circulante	1.003	1.255
Passivo não circulante	3.106	2.085

circulante líquido de $380.000. Se todo o ativo circulante fosse liquidado hoje, a empresa receberia $1,6 milhão. Qual é o valor contábil dos ativos da Klingon hoje? Qual é o valor de mercado?

6. **Cálculo de impostos [OA3]** A Renata Ltda. teve $236.000 de receita tributável em 2009. Usando as alíquotas da Tabela 2.3, calcule os impostos sobre a renda da empresa em 2009.

7. **Alíquotas tributárias [OA3]** No Problema 6, qual é a alíquota tributária média? Qual é a alíquota tributária marginal?

8. **Cálculo do FCO [OA4]** A Até Breve Ltda. tem vendas de $27.500, custos de $13.280, despesas de depreciação de $2.300 e despesas de juros de $1.105. Se a alíquota tributária for de 34%, qual é o FCO?

9. **Cálculo dos gastos líquidos de capital [OA4]** O balanço patrimonial de 2009 do Centro de Formação de Condutores Direção Perigosa Ltda. mostrou ativo não circulante líquido de $4,6 milhões, e o balanço patrimonial de 2008 mostrou ativo não circulante líquido de $5,2 milhões. A demonstração de resultados da empresa em 2009 mostrou uma despesa com depreciação de $385.000. Quais foram os gastos líquidos de capital em 2009?

10. **Cálculo dos acréscimos ao CCL [OA4]** O balanço patrimonial da João Pessoa S/A mostrou um ativo circulante de $2.100 e um passivo circulante de $1.380 em 2008. O balanço patrimonial em 2009 mostrou um ativo circulante de $2.250 e um passivo circulante de $1.710. Qual foi a variação do CCL da empresa em 2009?

11. **Fluxo de caixa para os credores [OA4]** Em 2008, o balanço patrimonial da Loja de Tênis da Maria S/A mostrou passivo não circulante de $2,6 milhões, e o balanço patrimonial em 2009 apresentou um passivo não circulante de $2,9 milhões. A demonstração de resultados de 2009 mostrou despesas com juros de $170.000. Qual foi o fluxo de caixa para os credores em 2009?

12. **Fluxo de caixa para os acionistas [OA4]** O balanço patrimonial de 2008 da Loja de Tênis da Maria S/A mostrou $740.000 na conta de capital social e $5,2 milhões na conta de ágio recebido na emissão de ações. O balanço patrimonial de 2009 apresentou $815.000 e $5,5 milhões nessas duas contas, respectivamente. Se a empresa pagou $490.000 em dividendos em 2009, qual foi o fluxo de caixa para os acionistas?

13. **Cálculo dos fluxos de caixa totais [OA4]** Com as informações da Loja de Tênis da Maria S/A apresentadas nos Problemas 11 e 12, suponhamos que você saiba também que os gastos líquidos de capital da empresa foram de $940.000 em 2009 e que ela reduziu seu investimento em capital circulante líquido em $85.000. Qual foi o seu fluxo de caixa operacional em 2009?

INTERMEDIÁRIO
(Questões 14-22)

14. **Cálculo dos fluxos de caixa totais [LO4]** A Nave Espacial Jetson S/A apresentou as seguintes informações em sua demonstração de resultados em 2009: vendas = $196.000; custos = $104.000; outras despesas = $6.800; despesas de depreciação = $9.100; despesas com juros = $14.800; impostos = $21.455; dividendos = $10.400. Além disso, a empresa emitiu $5.700 em novas ações em 2009 e resgatou $7.300 em dívidas de longo prazo.

    a. Qual foi o fluxo de caixa operacional em 2009?

    b. Qual foi o fluxo de caixa para os credores?

    c. Qual foi o fluxo de caixa para os acionistas?

    d. Se o ativo imobilizado líquido aumentou em $27.000 durante o ano, qual foi o acréscimo ao CCL?

15. **Uso das demonstrações de resultados [OA1]** Usando as seguintes informações da Pizzaria Rosatto Ltda., calcule as despesas de depreciação: vendas = $41.000; custos

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= $19.500; acréscimo aos lucros retidos = $5.100; dividendos distribuídos = $1.500; despesas com juros = $4.500; alíquota tributária = 34%.

16. Montagem de um balanço patrimonial [OA1] Monte um balanço patrimonial para a Bertinelli S/A com base nas seguintes informações: caixa = $195.000; patentes e direitos autorais = $780.000; contas a pagar = $405.000; contas a receber = $137.000; ativo imobilizado = $2.800.000; estoques = $264.000; títulos a pagar = $160.000; lucros retidos acumulados = $1.934.000; passivo não circulante = $1.195.300.

17. Direitos residuais [OA1] A Devolução S/A é obrigada a pagar $7.300 a seus credores durante o ano.

a. Qual é o valor de mercado do patrimônio dos acionistas se o ativo tiver um valor de mercado de $8.400?

b. E se o ativo for igual a $6.700?

18. Alíquotas tributárias marginais *versus* médias [OA3] (Consulte a Tabela 2.3.) A Companhia Crescimento tem $88.000 de lucros tributáveis e a Companhia Lucro tem $8.800.000 de lucros tributáveis.

a. Qual é a carga tributária de cada empresa?

b. Suponha que ambas tenham criado um projeto novo que aumentará o lucro tributável em $10.000. Qual é o valor de impostos adicionais que cada empresa pagará? Por que esse montante é o mesmo?

19. Lucro líquido e FCO [OA2] Durante 2009, a Sombrinha Chuvosa S/A teve vendas de $730.000. O custo de mercadorias vendidas, as despesas administrativas e de vendas e as despesas de depreciação foram $580.000, $105.000 e $135.000, respectivamente. Além disso, a empresa teve despesas com juros de $75.000 e alíquota tributária de 34%. (Ignore créditos tributários e provisões.)

a. Qual foi o lucro líquido da Chuvosa em 2009?

b. Qual foi seu fluxo de caixa operacional?

c. Explique seus resultados em (a) e (b).

20. Valores contábeis *versus* fluxos de caixa [OA4] No Problema 19, suponha que a Sombrinha Chuvosa S/A tenha pago $25.000 em dividendos. Isso é possível? Se as despesas com ativo imobilizado líquido e capital circulante líquido foram zero e se nenhuma emissão de ações ocorreu durante o ano, que informações você tem sobre o passivo não circulante?

21. Cálculo dos fluxos de caixa [OA2] As Indústrias Dahlia tiveram os seguintes resultados operacionais em 2009: vendas = $22.800; custo das mercadorias vendidas = $16.050; despesas de depreciação = $4.050; despesas com juros = $1.830; dividendos distribuídos = $1.300. No início do ano, o ativo imobilizado líquido era $13.650, o ativo circulante $4.800 e o passivo circulante $2.700. Ao final do ano, o ativo imobilizado líquido foi de $16.800, o ativo circulante $5.930 e o passivo circulante $3.150. A alíquota tributária foi 34%.

a. Qual é o lucro líquido em 2009?

b. Qual é o fluxo de caixa operacional em 2009?

c. Qual é o fluxo de caixa dos ativos em 2009? Isso é possível? Explique.

d. Se nenhuma dívida foi contraída durante o ano, qual foi o fluxo de caixa para os credores? Qual foi o fluxo de caixa para os acionistas? Explique e interprete os sinais positivos e negativos de suas respostas nas questões de (a) a (d).

22. Cálculo dos fluxos de caixa [OA4] Considere as seguintes demonstrações financeiras simplificadas para as Empresas Cabeça de Papagaio:

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EMPRESAS CABEÇA DE PAPAGAIO
Balanços patrimoniais parciais de 2008 e 2009

Ativo			Passivo e patrimônio líquido		
	2008	2009		2008	2009
Ativo circulante	$ 653	$ 707	Ativo imobilizado líquido	$ 261	$ 293
Passivo circulante	2.691	3.240	Passivo não circulante	1.422	1.512

EMPRESAS CABEÇA DE PAPAGAIO
Demonstração de resultados de 2009

Vendas	$8.280
Custos	3.861
Depreciação	738
Juros pagos	211

a. Qual é o patrimônio líquido de 2008 e 2009?

b. Qual é a variação do capital circulante líquido de 2009?

c. Em 2009, a Cabeça de Papagaio comprou $1.350 em ativo imobilizado novo. Quanto ela vendeu em ativo imobilizado? Qual é o fluxo de caixa dos ativos no ano? (A alíquota tributária é 34%.)

d. Em 2009, a Cabeça de Papagaio contraiu $270 em dívida nova de longo prazo. Qual o valor de dívidas de longo prazo que a empresa pagou durante o ano? Qual foi o fluxo de caixa para os credores?

DESAFIO
(Questões 23-26)

23. Ativo imobilizado e depreciação [OA4] No balanço patrimonial, a conta do ativo imobilizado líquido (AIL) é igual à conta do ativo imobilizado total (AIT) (que registra o custo de aquisição dos ativos imobilizados) menos a conta da depreciação acumulada (DA) (que registra a depreciação acumulada do ativo imobilizado). Sabendo que AIL = AIT − DA, mostre que a equação para gastos líquidos de capital, $AIL_{final} - AIL_{inicial} + D$ (onde D são as despesas de depreciação durante o ano), é igual a $AIT_{final} - AIT_{inicial}$.

24 Alíquotas tributárias [OA3] Consulte as alíquotas tributárias marginais da pessoa jurídica na Tabela 2.3.

a. Em sua opinião, por que a alíquota tributária marginal nos Estados Unidos salta de 34 para 39% a um lucro tributável de $100.001 e, em seguida, volta para 34% a um lucro tributável de $335.001?

b. Calcule a alíquota tributária média para uma corporação nos Estados Unidos com exatamente $335.001 de lucro tributável. Isso confirma sua explicação na parte (a)? Qual é a alíquota tributária média para uma corporação com exatamente $18.333.334? A mesma explicação vale para esse caso?

c. As alíquotas de 39 e 38% representam aquilo que é chamado de "bolha" de imposto. Suponhamos que o governo estadunidense queira diminuir o limite superior da faixa do imposto marginal de 39% de $335.000 para $200.000. Qual deveria ser a nova alíquota da bolha de 39%?

Use as seguintes informações para a Taco Mel S/A nos Problemas 25 e 26 (assuma que a alíquota tributária é 34%):

	2008	2009
Vendas	$ 7.233	$ 8.085
Depreciação	1.038	1.085
Custos de mercadorias vendidas	2.487	2.942
Outras despesas	591	515
Juros	485	579
Caixa	3.792	4.041
Contas a receber	5.021	5.892
Títulos de dívida de curto prazo	732	717
Exigível a longo prazo	12.700	15.435
Ativo imobilizado líquido	31.805	33.921
Contas a pagar	3.984	4.025
Estoques	8.927	9.555
Dividendos	882	1.011

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25. **Demonstrações financeiras [OA1]** Crie uma demonstração de resultados e um balanço patrimonial para a empresa em 2008 e 2009.

26. **Cálculo do fluxo de caixa [OA4]** Calcule o fluxo de caixa dos ativos, o fluxo de caixa para os credores e o fluxo de caixa para os acionistas em 2009.

MINICASO

Fluxos de caixa e demonstrações de resultados da companhia Pranchas Pôr do Sol Ltda.

A Pranchas Pôr do Sol é uma pequena empresa produtora de pranchas de surfe em Ubatuba. Zé Marca, seu fundador, é encarregado de projetar e vender as pranchas de surfe; porém sua experiência é em surfe, não em negócios. Por isso, os registros financeiros da empresa não estão em dia.

Os recursos para o investimento inicial da Pranchas Pôr do Sol foram fornecidos por Zé, seus amigos e seus familiares. Como o investimento inicial foi relativamente pequeno e a empresa tem fabricado pranchas de surfe apenas para sua própria loja, os investidores não exigiram que Zé tivesse demonstrações contábeis detalhadas. Porém, graças ao reconhecimento entre os surfistas profissionais, as vendas aumentaram recentemente, e Zé Marca está pensando numa expansão. Seus planos incluem a abertura de outra loja de pranchas de surfe em Fernando de Noronha, bem como o fornecimento de suas "sticks" (pranchas, na gíria de alguns surfistas) para outros vendedores.

Os planos de expansão do Zé Marca exigem um investimento significativo, que ele pretende financiar com uma combinação de recursos extras de fontes externas mais algum dinheiro emprestado de bancos. Naturalmente, os novos investidores e credores exigem demonstrações financeiras mais organizadas e detalhadas do que aquelas que ele montava anteriormente. Atendendo às exigências de seus investidores, Zé contratou a analista financeira Cristina Lobos para avaliar o desempenho da empresa no último ano.

Após examinar extratos bancários antigos, recibos de vendas, declarações de imposto de renda e outros registros, Cristina reuniu as seguintes informações:

	2008	2009
Custos de produtos vendidos	$126.038	$159.143
Caixa	18.187	27.478
Depreciação	35.581	40.217
Despesas com juros	7.735	8.866
Despesas administrativas e de vendas	24.787	32.352
Contas a pagar	32.143	36.404
Ativo imobilizado líquido	156.975	191.250
Vendas	247.259	301.392
Contas a receber	12.887	16.717
Títulos a pagar	14.651	15.997
Dívidas de longo prazo	79.235	91.195
Estoques	27.119	37.216
Capital social novo	0	15.600

A Pranchas Pôr do Sol, no momento, distribui 50% do lucro líquido como dividendos para Zé Marca e para os outros investidores originais, e tem uma alíquota tributária total de 20%. Você é assistente de Cristina e ela pediu para que você preparasse o seguinte:

1. Uma demonstração de resultados de 2008 e 2009.
2. Um balanço patrimonial de 2008 e 2009.
3. O fluxo de caixa operacional de cada ano.
4. O fluxo de caixa dos ativos em 2009.
5. O fluxo de caixa para os credores em 2009.
6. O fluxo de caixa para os acionistas em 2009.

QUESTÕES

1. Como você descreveria os fluxos de caixa da Pranchas Pôr do Sol em 2009? Escreva um resumo.
2. De acordo com o resumo feito na questão anterior, o que você acha dos planos de expansão de Zé Marca?

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